Higher Topos Theory - Self Study

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Contents

Abstract

I am currently reading Higher Topos Theory to understand the theory of quasi-categories and basic ∞-category theory. My goal is to understand the generalizations of basic 1-categorical concepts to the ∞-categorical setting, such as (co)limits, adjunctions, and Yoneda's lemma. I'm also interested in more advanced topics in higher category theory, most of which require a basic understanding of the theory of ∞-categories as a prerequisite.

I began reading this text on August 8th, 2025, at 11:06 PM, as an independent project. I am hoping to be able to find a professor who is willing to support my readings, but I am unsure that will happen. Nevertheless, I intend to gain a working understanding of ∞-categories over the next few months, and a more advanced understanding come the spring semester. Since this text is quite advanced and is not meant as a textbook, but rather as a reference, I will be taking extremely detailed and extensive notes on the text. I will also be referencing supplementary resources throughout my reading, so I have decided it worthwhile to maintain a bibliography for these notes. Any notes not credited to a bibliography source are either from [Lur09], or directly from me.

Chapter 1

An Overview of Higher Category Theory

The first chapter introduces quasi-categories as models for ∞-categories, and then develops ∞-categorical analogues of standard 1-categorical notions.

1.1 Foundations of Higher Category Theory

Terminology 1.1.0.1. Throuhgout this paper, the term *space* is short for a topological space, and almost always a compactly generated topological space. The *homotopy type* of a topological space is a choice of a weak homotopy equivalent topological space, or equivalently the collection of weak homotopy equivalent spaces.

1.1.1 Goals and Obstacles

The theory of ∞-categories finds natural motivation in the fundamental n-groupoids of a topological space X. Define $\Pi_n(X)$ to take points $x \in X$ as objects, paths from x to y as 1-morphisms, homotopies of paths as 2-morphisms, and so on until n. Since vertical composition of homotopies is only associative up to higher homotopy, we are forced to take n-morphisms to be *homotopy classes*, rather than simply homotopies. Of course $\Pi_n(X)$ is a groupoid because all paths and homotopies admit inverses.

One might hope that we can fix this problem by considering an ∞-groupoid, or an $(\infty, 0)$-category, which would not take homotopies as n-morphisms for all $n \geq 2$. Interestingly, the ∞-groupoid $\Pi_\infty(X)$ of a space X encodes the homotopy type of X, meaning the study of ∞-groupoids simplifies to the study of spaces in classical homotopy theory, and we have many ways of describing homotopy types of spaces. It is also a commonly accepted principle of higher category theory that all $(\infty, 0)$-categories are of the form $\Pi_\infty(X)$ for some space X. This principle is profound, since it gives us a place to begin our work in higher category theory.

In general, an n-category should be a category enriched over $(n-1)$-categories. However, for finite n, this process produces the notion of a *strict* n-category, and n-category theory is more naturally viewed via the theory of *weak* n-categories. These issues make the theory of n-categories fairly difficult to work with, but they vanish when one passes to ∞-categories. An (∞, n)-category can indeed be viewed as a category enriched over $(\infty, n-1)$-categories. This discussion suggests a definition (or more precisely, a blueprint) for ∞-categories. An ∞-category $\mathcal{C}$ should consist of objects $\mathrm{Obj}(\mathcal{C})$, and for each $x, y \in \mathcal{C}$, a ∞-groupoid $\mathrm{Map}_{\mathcal{C}}(x, y)$ with an associative composition law. The theory of ∞-categories does not care whether this law be strictly associative or only associative up to coherent homotopy, since any ∞-category with coherently associative multiplication is equivalent to one with strictly associative multiplication.

Definition 1.1.1.1. A *topological category* is a category enriched over $\mathcal{CG}$, the category of compactly generated spaces. The category of topological spaces will be denoted by $\mathcal{C}\mathrm{at}_{\mathrm{top}}$.

We recall from our knowledge of enriched category theory that this implies a topological category $\mathcal{C}$ has topological spaces $\mathrm{Map}_{\mathcal{C}}(x, y)$ for each pair $x, y \in \mathcal{C}$, with an associative composotion law

$$\mathrm{Map}_{\mathcal{C}}(y, z) \times \mathrm{Map}_{\mathcal{C}}(x, y) \to \mathrm{Map}(x, z),$$

where the product is computed in $\mathcal{CG}$.

Remark 1.1.1.2. As is often the case, working in $\mathcal{CG}$ is done largely as a technical point which can be ignored.

1.1.2 ∞-Categories

Definition 1.1.1.1 is the most obvious definition of an ∞-category, but it is also one of the hardest to actually work with. Our use of topological categories in this book will be more as a "benchmark" of sorts. Any model for ∞-categories which is equivalent to that of topological categories must necessarily be a good one. Many better models exist, and in this text we choose to use the theory of *weak Kan complexes*. These were first introduced by Boardman and Vogt in [BV73], and were more recently and more extensively studied by André Joyal in [Joy08a, Joy08b], who instead called them *quasi-categories*, which the modern literature now uses. However, Lurie's book almost exclusively uses quasi-categories as our model for ∞-categories, so for the purposes of these notes we will use the terms "quasi-category" and "∞-category" more or less interchangeably.

Quasi-categories are built atop the theory of *simplicial sets*, and any readers of HTT (or these notes) should have at least an elementary understanding of the subject. I recommend my own exposition [Tal25], which builds the abstract theory of simplicial sets without forsaking geometric intuition, but also without dwelling excessively on the topological impliciations of the theory. We also recommend [Rie11, Fri23]. The former should prepare the reader for much of the theory in this paper, the rest of which can be picked up along the way, although we feel Riehl strays too far from the geometric intuition of simplices. Conversely, Friedman's introduction is well-liked for the geometric intuition it provides, but we feel it focuses too much on topological aspects of the theory to be the best preparation for the material covered in HTT.

The theory of simplicial sets is of importance due to the homotopic aspects of the theory. Recall from our discussion in section 1.1.1 that the ∞-groupoid $\Pi_\infty(X)$ of a space X captures the homotopy type of X. We also know from [Qui67] that since $|-| \dashv \mathrm{Sing}$ is a Quillen equivalence, where Sing is the singular complex functor. This means in particular that the unit and counit of the adjunction are natural weak homotopy equivalences, meaning spaces are determined by their singular sets up to homotopy type. Thus, if we are only interested in the study of spaces up to weak homotopy equivalence, we might as well work in the category of simplicial sets. We might also think that ∞-groupoids should be able to be modeled by singular sets, since they both encode the homotopy type of a space. Indeed this is possible, but we prefer to work slightly more generally. Singular sets satsify the following important property.

Definition 1.1.2.1. Let K be a simplicial set. We say K is an *Kan complex* if, for all $n \geq 0$ and for all $0 \leq i \leq n$, any diagram of solid arrows

$$\begin{tikzcd} \Lambda^n_i \arrow[r] \arrow[d, hook] & X \\ \Delta^n \arrow[ur, dotted] & \end{tikzcd}$$

admits a (not necessarily unique) dotted arrow as indicated, rendering the diagram commutative.

Recall here that $\Lambda^n_i \subseteq \Delta^n$ represents the ith horn of the standard n-simplex; we defer the reader to [Tal25, §3.4] for a rigorous definition.

Remark 1.1.2.2. The singular complex of a topological space is a Kan complex. Showing this first involves showing that $|\Lambda^n_i|$ is a retract of $|\Delta^n|$ in $\mathcal{T}\mathrm{op}$. There also exist combinatorial approaches to extracting homotopy groups from any Kan complex K, which agree with the homotopy groups on $|K|$. Thus, Kan complexes behave like the homotopy type of a space.

Simplicial sets, and in particular Kan complexes, thus should form a model for ∞-groupoids, which we note are an important special case of ∞-categories. Indeed, Kan complexes are a common model for ∞-groupoids, which we will explore later. However, another special case of ∞-categories that our definition must reflect is the notion of a basic 1-category. If an ∞-category $\mathcal{C}$ has no nontrivial n-morphisms for $n > 1$, then clearly $\mathcal{C}$ can be regarded as a 1-category. Luckily, simplicial sets are also closely related to category theory. Recall the nerve $N(\mathcal{C})$ of a category is a simplicial set whose n-simplices are functors $[n] \to \mathcal{C}$, which carry the data of an n-tuple of composable morphisms.

Remark 1.1.2.3. The nerve encodes the full information of a category, and we can in fact recover the initial category $\mathcal{C}$ from the nerve $N(\mathcal{C})$ with the following list of observations.

1. The 0-simplices $N(\mathcal{C})_0$ are the objects, so we define $\mathrm{Obj}(\mathcal{C}) = N(\mathcal{C})_0$.
2. The 1-simplices $N(\mathcal{C})_1$ are simply morphisms, so we define $\mathrm{Mor}(\mathcal{C}) = N(\mathcal{C})_1$.
3. Given an object $x \in \mathcal{C}$, the degeneracy $s_0 : N(\mathcal{C})_0 \to N(\mathcal{C})_1$ maps to the identity $x \mapsto 1_x$. Thus, the identity function $x \mapsto 1_x$ is simply the degeneracy $s_0 = 1_\bullet$.
4. Given a morphism $f \in N(\mathcal{C})_1$, the 0th face $d_0 : N(\mathcal{C})_1 \to N(\mathcal{C})_0$ maps to the codomain $f \mapsto \mathrm{cod}\, f$, and similarly the first face $d_1 : N(\mathcal{C})_1 \to N(\mathcal{C})_0$ maps to the domain $f \mapsto \mathrm{dom}\, f$. We may thus define $\mathrm{dom}, \mathrm{cod} : \mathrm{Mor}(\mathcal{C}) \to \mathrm{Obj}(\mathcal{C})$ as the faces $\mathrm{dom} = d_1$ and $\mathrm{cod} = d_0$.
5. A composable pair (g, f) is a 2-simplex $(g, f) \in N(\mathcal{C})_2$. The 1st face $d_1 : N(\mathcal{C})_2 \to N(\mathcal{C})_1$ maps composable pairs to their composite $(g, f) \mapsto gf$. Thus, composition is simply the 1st face $d_1 = \circ$.
6. Composites of larger tuples of maps arise from higher degeneracies, and associativity follows from the simplicial identities. The unit law for identities under composition also follows from simplicial identities.

A more explicit description of remark 1.1.2.3 (5,6) will be helpful later. A morphism from x to y is defined as a 1-simplex $f \in N(\mathcal{C})_1$ with $d_1(f) = x$ and $d_0(f) = y$. Given two morphisms $x \xrightarrow{f} y \xrightarrow{g} z$, the composite $g \circ f$ can be characterized by noting there exists a 2-simplex $(g, f) = \sigma \in N(\mathcal{C})_2$ such that

1. $d_0(\sigma) = g$;
2. $d_1(\sigma) = gf$;

3. $d_2(\sigma) = f$.

We say σ *witnesses* gf *as a composite of* g *and* f.

There exists a characterization of simplicial sets that arise as nerves, which is more straightforward than the characterization of Kan complexes.

Notation 1.1.2.4. Let J be a linearly ordered set. Denote by Δ^J the representable functor $\mathrm{Hom}(-, J)$ in the *category of linearly ordered sets*, restricted along the objects of $\mathbf{\Delta}$. Clearly, $\Delta^{[n]} = \Delta^n$, so this is simply a mild generalization of notation we already had.

Proposition 1.1.2.5. *Let* K *be a simplicial set. The following conditions are equivalent.*

1. *There exists a small category* $\mathcal{C}$ *and an isomorphism* $K \cong N(\mathcal{C})$.
2. *For each* $0 < i < n$, *and each diagram of solid arrows*

$$\begin{tikzcd} \Lambda^n_i \arrow[r] \arrow[d, hook] & K \\ \Delta^n \arrow[ur, dashed] & \end{tikzcd}$$

there is a unique *dashed arrow as indicated, rendering the diagram commutative.*

Proof. We mostly follow Lurie's proof in [Lur09, Prop. 1.1.2.2], but we take a different route for showing agreement for (1), and Lurie assumes a slightly more advanced understanding of simplicial sets than I currently have, so at some points in this proof we may prove extra facts. For example, we start by proving that for $0 < i < n$, the vertices of the ith horn are $(\Lambda^n_i)_0 = [n]$. Note that for i to exist, we must have $n \geq 2$. For $n = 2$, there exist at two coface maps $d^j : \Delta^1 \to \Delta^2$ with $j \neq i$. 0-simplices of Δ^1 are arrows $[0] \to [1]$, which correspond to elements of $[1]$. More precisely, arrows are uniquely determined by their image. Thus, 0-simplices in the face $\partial_j \Delta^n_0 = \operatorname{im} d^j_0$ will also be uniquely determined by their image, so it suffices to show each $x \in [2]$ lands in the image of some coface d^j with $j \neq i$. This is immediate: since d^j only skips j, and since another coface d^k, $k \neq i$ exists which doesn't skip j, every element shows up. Inducting on this same logic proves the statement for $n \geq 2$, and vacuously for all n.

Now we show for $0 < k \leq n$ and $n \geq 2$, there is a simplicial subset $\Delta^{\{k-1,k\}} \subseteq \Lambda^n_i$. Recall that m-simplices f in the horn are composites $[m] \xrightarrow{f} [n-1] \xrightarrow{d^{j \neq i}} [n]$. Let $g : [m] \to \{k-1, k\} \subseteq [n]$ be an m-simplex in $\Delta^{k-1,k}$. If $k < n$, then $k, k-1 \in [n-1]$. Write g' for the map with codomain viewed as a subset $\{k-1, k\} \subseteq [n-1]$. Since $i \neq n$, we have that $\operatorname{im} d^n \subseteq \Lambda^n_i$. Thus, $d^n g' = g \in \operatorname{im} d^n_m$, and thus is an m-simplex in the horn. If $k = n$, then since $n \geq 2$, we have $k - 1 > 0$. Since $i \neq 0$, we do the same argument.

If $n < 2$ then the statement is true vacuously, so let $n \geq 2$. Let K be the nerve of a small category $\mathcal{C}$. Let $f_0 : \Lambda^n_i \to K$ with $0 < i < n$, and let $x_k \in \mathcal{C}$ denote the image of the vertex $k \in (\Lambda^n_i)_0 = [n]$ under f_0. Choose some $0 < k \leq n$, and consider the restriction $f_0|\Delta^{\{k-1,k\}}$. Note that the 1-simplices of $\Delta^{\{k-1,k\}}$ are

$$\Delta^{\{k-1,k\}}_1 = \left\{ (0,1) \mapsto k, (0,1) \mapsto k-1, \begin{matrix} 0 \mapsto k-1 \\ 1 \mapsto k \end{matrix} \right\}.$$

The first two are easily seen to be the degenerates $s_0(k), s_0(k-1)$, respectively. Since f_0 is a morphism of simplicial sets, $f_0 s_0(k) = s_0 f_0(k) = s_0(x_k) = 1_{x_k}$. This is uninteresting, so let g_k denote the unique nondegenerate simplex. Note that the image under f is a 1-simplex in $N(\mathcal{C})$; that is a morphism $g_k : c \to d$. We want to show that with notation as above, $c = x_{k-1}$ and $d = x_k$. Indeed, note that for 1-simplices in the nerve, the first face is simply *domain* and the 0th face is *codomain*. Since f_0 commutes with faces, we have

$$\operatorname{dom} g_k = d_1 f_0(g_k) = f_0 d_1(g_k) = f_0(g_k \circ d^1) = f_0(0 \mapsto 0 \mapsto k-1) = x_{k-1},$$

since $0 \mapsto 0 \mapsto k-1 = g_k \circ d^1$ is a 0-simplex, which is uniquely determined (and identified with) its image. The same follows for domain.

We thus have for each $0 < k \leq n$, there is an arrow $g_k : x_{k-1} \to x_k$, allowing us to form a composable chain

$$x_0 \xrightarrow{g_1} x_1 \xrightarrow{g_2} \cdots \xrightarrow{g_n} x_n.$$

This is an n-simplex in Δ^n. We can similarly construct m-simplices for $m \leq n$ by requiring $k \leq m$. We claim that this relation allows us to define a mapping $f : \Delta^n \to K$.

To see how this follows, we invoke the Yoneda lemma to recall that there is an isomorphism $\mathcal{S}\mathrm{et}^{\mathbf{\Delta}^{\mathrm{op}}}(\Delta^n, K) \cong K_n$ given by evaluating the nth component of natural transformations at the identity. In particular, Yoneda says a morphism $f : \Delta^n \to K$ is determined by $f_n(1_{[n]})$. We choose $f_n(1_{[n]})$ to be the n-simplex $(g_1, \ldots, g_n)$ of $N(\mathcal{C})$ considered above. The remainder of the proof will focus on showing this definition indeed agrees with Λ^n_i. For now, we note that f is manifestly unique, since we obtained a unique composite constructed in this way, so any other map $\Delta^n \to K$ which restricts along the horn to f_0 must necessarily correspond to the same chain of morphisms.

We now begin to show $f|\Lambda^n_i = f_0$. First, we show they agree on cofaces. Let $d^j : [n-1] \to [n]$ be an $(n-1)$-simplex in Λ^n_i. Then since f is a morphism of simplicial sets and commutes with faces, $d^j = 1_{[n]} d^j = d_j(1_{[n]})$ implies that $f(d^j) = d_j f(1_{[n]})$. For simplicity, we take d^j to be an inner face $0 < j < n$, and simply note the argument for outer faces is the same. From our knowledge of the nerve, we have that

$$f(d^j) = f d_j(1_{[n]}) = d_j f(1_{[n]}) = d_j \left(x_0 \xrightarrow{g_1} \cdots \xrightarrow{g_n} x_n \right)$$

$$= x_0 \xrightarrow{g_1} \cdots \xrightarrow{g_{j-1}} x_{j-1} \xrightarrow{g_{j+1} g_j} x_{j+1} \xrightarrow{g_{j+1}} \cdots \xrightarrow{g_n} x_n$$

It suffices to show $f_0(d^j)$ agrees with this composite. One can do this by showing each object and morphism align with that of $f(d^j)$, which involves an induction argument that is extremely lengthy due to needing to show many cases, but is in general very simple. We trust the reader to be able to do this argument, and we also trust that the reader knows I can do the argument too and just don't want to, so we prefer not to spend 2 pages on it and instead continue with the proof.

Now, we show this agreement on cofaces implies f and f_0 agree in general. We note that every m-simplex $h \in \Delta^n_m$ is a morphism $h : [m] \to [n]$ in $\mathbf{\Delta}$, which admits a factorization into cofaces and codegeneracies $h = d^{j_1} \circ \cdots d^{j_\ell} \circ s^{p_1} \circ \cdots \circ s^{p_q}$. Since faces and degeneracies in Δ^n are given by precomposing by the corresponding cofaces and codegeneracies in $\mathbf{\Delta}$, we have

$$h = d^{j_1} \circ \cdots \circ s^{p_q} = 1_{[n]} \circ d^{j_1} \circ \cdots \circ s^{p_q} = (s_{p_q} \circ \cdots \circ d_{j_1})(1_{[n]}).$$

Since f is a morphism of simplicial sets, we have $f s_j = s_j f$, and likewise for faces. Induction gives

$$f(h) = (f \circ s_{p_q} \circ \cdots \circ d_{j_1})(1_{[n]}) = (s_{p_q} \circ \cdots \circ d_{j_1})(f(1_{[n]})) = (s_{p_q} \circ \cdots \circ d_{j_2})(f(d_{j_1}(1_{[n]}))).$$

Now we want to show f agrees with f_0 on m-simplices of Λ^n_i. Since $h \in (\Lambda^n_i)_m$, there is some m-simplex h' such that $h = d^j h'$ for $j \neq i$, so by uniqueness we have that $d^{j-1} \neq d^i$ in the epi-mono factorization of h. Since f and f_0 agree on cofaces,

$$\begin{aligned} f(h) &= (s_{p_q} \circ \cdots \circ d_{j_2})(f(d_{j_1}(1_{[n]}))) \\ &= (s_{p_q} \circ \cdots \circ d_{j_2})(f(d_{j_1})) \\ &= (s_{p_q} \circ \cdots \circ d_{j_2})(f_0(d_{j_1})) \\ &= (s_{p_q} \circ \cdots \circ d_{j_2})(f_0(d_{j_1}(1_{[n]}))) \\ &\overset{!}{=} f_0(h), \end{aligned}$$

where the equality marked by ! follows since f_0 is a morphism of simplicial sets and commutes with faces and degeneracies. Thus, the statement ($1 \implies 2$) is proven.

Now for the converse. Let K satisfy (2), and define a category $\mathcal{C}$ as follows.

1. Objects are vertices: $\mathrm{Obj}(\mathcal{C}) = K_0$;
2. Morphisms are edges: $\mathrm{Mor}(\mathcal{C}) = K_1$. Domain and codomain are faces: $\mathrm{dom} = d_1$ and $\mathrm{cod} = d_0$;
3. Identity $1_\bullet : \mathrm{Obj}(\mathcal{C}) \to \mathrm{Mor}(\mathcal{C})$ is the degeneracy $1_\bullet = s_0$;
4. Let $f : x \to y$ and $g : y \to z$ in $\mathcal{C}$. We want to define a morphism $\sigma_0 : \Lambda^2_1 \to K$. Define the action on 0-simplices by $0 \mapsto x$, $1 \mapsto y$, and $2 \mapsto z$. Denote by g_k the nondegenerate simplex in $\Delta^{k-1,k}$ for $0 < k \leq 2$. Then define the map as $g_1 \mapsto f$ and $g_2 \mapsto g$. We will show later that this data determines a morphism $\sigma_0 : \Lambda^2_1 \to K$. By the horn filling condition, this extends to a map $\sigma : \Delta^2 \to K$. Let $d^1 : [1] \to [2]$ be the unique 1-simplex of Δ^2 with $d^1(0) = 0$ and $d^1(1) = 2$. Then define $g \circ f$ as $\sigma(d^1)$.

Of course, we need to show the data in (4) defines a morphism $\sigma_0 : \Lambda^2_1 \to K$. Recall that $\Lambda^2_1 = \partial_0 \Delta^2 \cup \partial_2 \Delta^2$. By universality of union, if we can define maps $\partial_0 \Delta^2 \to K$ for $i = 0, 2$, such that these maps agree on $\partial_0 \Delta^2 \cap \partial_2 \Delta^2$, then they assemble into a map Λ^2_1. We may thus construct a map in this way using only the data of (4). Recall that $\partial_i \Delta^n = \mathrm{im}(d^i : \Delta^{n-1} \to \Delta^n)$, and in particular $d^i : \Delta^{n-1} \to \partial_i \Delta^n$ is an isomorphism by uniqueness of the epi-mono factorization in $\mathbf{\Delta}$, with a well-defined inverse $d^i \circ h \mapsto h$. Thus, if $\tau_0 : \Delta^{n-1} \to K$ is a morphism of simplicial sets, then there is a morphism of simplicial sets $\tau : \partial_i \Delta^n \to K$ given by $d^i \circ h \mapsto h \mapsto f_0(h)$, since this is simply composition of the inverse of the isomorphism with the map f_0. In particular, maps $\Delta^1 \to K$ induce maps $\partial_i \Delta^2 \to K$.

By the Yoneda lemma, maps $\tau_0' : \Delta^1 \to K$ are specified by the image $\tau_0'(1_{[1]})$. Thus, the map $\tau_0 : \partial_i \Delta^2 \to K$ is fully specified by the image $\tau_0'(1_{[1]})$ in the aforementioned manner. Define

$$\alpha_0 : \Delta^1 \to K, \qquad 1_{[1]} \mapsto f,$$

$$\beta_0 : \Delta^1 \to K, \qquad 1_{[1]} \mapsto g.$$

These of course assemble into maps $\alpha : \partial_2 \Delta^2 \to K$ and $\beta : \partial_0 \Delta^2 \to K$. To show these maps agree on the intersection, it suffices to show they agree on the vertex $1 \in \partial_{0,2} \Delta^2_0$. Indeed, $1 \in \partial_0 \Delta^2_0$ is simply $d^0(0)$, and $1 \in \partial_2 \Delta^2_0$ is simply $d^2(1)$. Thus, they are mapped

$$\alpha : 1 = d^2(1) \mapsto 1 \mapsto \alpha_0(1), \qquad \beta : 1 = d^0(0) \mapsto 0 \mapsto \beta_0(0).$$

We simply need to show these agree. Since β_0 and α_0 are morphisms of simplicial sets, they necessarily commute with faces, meaning the diagrams

$$\begin{tikzcd} \Delta^1_1 \arrow[r, "\alpha_0"] \arrow[d, "d_0"'] & K_1 \arrow[d, "d_0"] \\ \Delta^1_0 \arrow[r, "\alpha_0"'] & K_0 \end{tikzcd} \qquad \begin{tikzcd} \Delta^1_1 \arrow[r, "\beta_0"] \arrow[d, "d_1"'] & K_1 \arrow[d, "d_1"] \\ \Delta^1_0 \arrow[r, "\beta_0"'] & K_0 \end{tikzcd}$$

commute. Recall that $\alpha_0(1_{[1]}) = f$ and $\beta_0(1_{[1]}) = g$. Following the identity around both squares yields the equalities

$$\begin{tikzcd} 1_{[1]} \arrow[r, mapsto, "\alpha_0"] \arrow[d, mapsto, "d_0"'] & f \arrow[d, mapsto, "d_0"] \\ 1_{[1]} \circ d^0 = 1 \arrow[r, mapsto, "\alpha_0"'] & \alpha_0(1) = d_0(f) \end{tikzcd} \qquad \begin{tikzcd} 1_{[1]} \arrow[r, mapsto, "\beta_0"] \arrow[d, mapsto, "d_1"'] & g \arrow[d, mapsto, "d_1"] \\ 1_{[1]} \circ d^1 = 0 \arrow[r, mapsto, "\beta_0"'] & \alpha_0(0) = d_1(g) \end{tikzcd}$$

Recall that in (2), we defined $d_1(g) = \operatorname{dom} g$ and $d_0(f) = \operatorname{cod} f$. In (4), we defined $\operatorname{cod} f = y = \operatorname{dom} g$, and thus the morphisms agree. We thus have an induced map $\sigma_0 = \alpha \cup \beta : \Lambda^2_1 \to K$, which is immediately guarenteed to be defined as claimed in (4).

Now that our definitions are indeed defined, we must show $\mathcal{C}$ defines a category, as claimed.

1. First, we show the identity is indeed an identity. Let $y \in \mathcal{C}$, and let $f : x \to y$ and $g : y \to z$ be morphisms in $\mathcal{C}$. The data $(f, 1_y)$ consitutes a horn $x \xrightarrow{f} y \xrightarrow{1} y$, which fills to a 2-simplex $\tau \in K_2$ and a morphism $\sigma : \Delta^2 \to K$ given by $1_{[2]} \mapsto \tau$, such that τ witnesses $1_y \circ f$ as the composite of 1_y and f, as described in (4). We show $1_y \circ f = f$. Recall $1_y \circ f = \sigma(d^1)$. We have
$$\begin{aligned} d_0(\tau) &= d_0\sigma(1_{[2]}) = \sigma d_0(1_{[2]}) = \sigma(d^0) = 1_y = s_0(y), \\ d_1(\tau) &= d_1\sigma(1_{[2]}) = \sigma d_1(1_{[2]}) = \sigma(d^1) = 1_y \circ f, \\ d_2(\tau) &= d_2\sigma(1_{[2]}) = \sigma d_2(1_{[2]}) = \sigma(d^2) = f. \end{aligned}$$
Recall that there is a simplicial identity $s_i d_i = s_i d_{i+1} = 1$. We have
$$s_1(f) = s_1 d_2(\tau) = \tau,$$
and thus τ is degenerate. We also have
$$1_y \circ f = d_1(\tau) = d_1 s_1(f) = f$$
by the same simplicial identity. The converse direction is the same argument, except we find that $s_0(g) = \tau$.

2. Now we show associativity. Let
$$w \xrightarrow{f} x \xrightarrow{g} y \xrightarrow{h} z$$
be a composable triple in $\mathcal{C}$. We want to show $h(gf) = (hg)f$. Define the following 2-simplices of K to be the images of $1_{[2]}$ under the relevant fillers:
$$\sigma_{gf} = \begin{cases} d_0(\sigma_{gf}) = g, \\ d_1(\sigma_{gf}) = gf, \\ d_2(\sigma_{gf}) = f \end{cases} \qquad \sigma_{hg} = \begin{cases} d_0(\sigma_{hg}) = h, \\ d_1(\sigma_{hg}) = hg, \\ d_2(\sigma_{hg}) = g \end{cases}$$

$$\sigma_{h(gf)} = \begin{cases} d_0(\sigma_{h(gf)}) = h, \\ d_1(\sigma_{h(gf)}) = h(gf), \\ d_2(\sigma_{h(gf)}) = gf \end{cases} \qquad \sigma_{(hg)f} = \begin{cases} d_0(\sigma_{(hg)f}) = hg, \\ d_1(\sigma_{(hg)f}) = (hg)f, \\ d_2(\sigma_{(hg)f}) = f \end{cases}$$

By the same logic as earlier (and indeed, we will show this holds generally later with an inductive argument), there is a horn $\tau_0 : \Lambda^3_2 \to K$ and a filler τ defining the 3-simplex $\lambda = \tau(1_{[3]}) \in K_3$. This simplex has faces

$$d_0(\lambda) = d_0\tau(1) = \tau(d^0) = \sigma_{hg},$$

$$d_1(\lambda) = d_1\tau(1) = \tau(d^1) = \sigma_{h(gf)},$$

$$d_3(\lambda) = d_3\tau(1) = \tau(d^3) = \sigma_{gf}.$$

We first want to show $d_2(\lambda) = \sigma_{(hg)f}$. For now, we note that

$$d_0 d_2(\lambda) = d_1 d_0(\lambda) = d_1(\sigma_{hg}) = hg,$$

$$d_2 d_2(\lambda) = d_2 d_3(\lambda) = d_2(\sigma_{gf}) = f,$$

by the simplicial identity $d_i d_j = d_{j-1} d_i$ for $i < j$. This defines a horn, which admits a filler. Since $\sigma_{(hg)f}$ agrees with the horn but is a 2-simplex, it necessarily is a filler, and by (2) this filler must be unique. Thus, $d_2(\lambda) = \sigma_{(hg)f}$ is the unique filler. Finally, note that by the same simplicial identity,

$$(hg)f = d_1 d_2(\lambda) = d_1 d_1(\lambda) = h(gf).$$

Therefore, $\mathcal{C}$ is a category, as claimed.

Finally, we show the isomorphism $K \cong N(\mathcal{C})$. By construction, we have a map $\varphi : K \to N(\mathcal{C})$ given by mapping n-simplices to the n-fold composite they witness. Of course, this is a morphism of simplicial sets basically by construction, which is checked by a simple (but annoying to type up) proof. We simply show φ is an isomorphism. It suffices to show it is an isomorphism levelwise, meaning $\varphi_n : K_n \cong N(\mathcal{C})_n$. By the Yoneda lemma, this is equivalent to showing isomorphism

$$\mathrm{s}\mathcal{S}\mathrm{et}(\Delta^n, K) \cong \mathrm{s}\mathcal{S}\mathrm{et}(\Delta^n, N(\mathcal{C}))$$

for all $n \geq 0$.

We induct on this statement. This is obvious by construction for $n = 0$ and $n = 1$, since $N(\mathcal{C})_i = K_i$ when $i = 0, 1$, so assume the inductive hypothesis for $n \geq 2$, and choose $0 < i < n$ an integer. Precomposition by the canonical inclusion $\Lambda^n_i \hookrightarrow \Delta^n$ and postcomposition by φ induce a commutative diagram

$$\begin{tikzcd} \mathrm{s}\mathcal{S}\mathrm{et}(\Delta^n, K) \arrow[r] \arrow[d] & \mathrm{s}\mathcal{S}\mathrm{et}(\Delta^n, N(\mathcal{C})) \arrow[d] \\ \mathrm{s}\mathcal{S}\mathrm{et}(\Lambda^n_i, K) \arrow[r] & \mathrm{s}\mathcal{S}\mathrm{et}(\Lambda^n_i, N(\mathcal{C})) \end{tikzcd}$$

Recall by our first part of the proof that $N(\mathcal{C})$ satisfies property (2). Thus, since K and $N(\mathcal{C})$ satisfy (2), the vertical maps are bijective. It suffices to thus show the bottom map is bijective since isomorphisms satisfy 2-out-of-3. By the inductive hypothesis, there is an isomorphism $\mathrm{s}\mathcal{S}\mathrm{et}(\Delta^{n-1}, K) \cong \mathrm{s}\mathcal{S}\mathrm{et}(\Delta^{n-1}, N(\mathcal{C}))$. Recall from earlier that any face $\partial_j \Delta^n$ is isomorphic to Δ^{n-1}, and that the horn decomposes as maps for faces. Since the faces are isomorphic to Δ^{n-1} for which the statement follows, the statement must follow for each face, and by universality the statement follows for the horn, completing the proof. ■

Remark 1.1.2.6. We may refer to n-simplices of a simplicial set K as morphisms $\sigma : \Delta^n \to K$. We recall by our use of Yoneda in the above proof that such a morphism amounts to specifying an n-simplex $\sigma \in K_n$ and an image $1_{[2]} \mapsto \sigma$, justifying this terminology.

Remark 1.1.2.7. Of course, proposition 1.1.2.5 (2) is similar to definition 1.1.2.1, but we require it only for inner horns, and we require the filler be unique. It is fairly easy to see that it would be unreasonable to assume outer horns have unique fillers in proposition 1.1.2.5. For example, let $n = 2$. A horn $\Lambda^2_2 \to N(\mathcal{C})$ is a diagram of solid arrows

$$\begin{tikzcd} & y \arrow[dr] & \\ x \arrow[ur, dotted] \arrow[rr] & & z \end{tikzcd}$$

and a filler is a dotted arrow as indicated, rendering the diagram commutative. This is of course unreasonable to ask for, but it is reasonable for *groupoids*.

Simplicial sets are thus good models for ∞-groupoids when K is a Kan complex, and are good models for regular categories when K satisfies proposition 1.1.2.5 (2). We earlier noted that ∞-groupoids and regular 1-categories are both special cases of ∞-categories, so we would expect some more general class of simplicial sets to give us a model. To see how this arises, we use our observations from proposition 1.1.2.5 and remark 1.1.2.3 to consider how a simplicial set K should be interpreted to yield an ∞-category.

First, clearly objects are vertices, morphisms are edges, and identity, domain, and codomain are the same as remark 1.1.2.3. A 2-simplex $\sigma : \Delta^2 \to K$ is thought of as a diagram

$$\begin{tikzcd} & y \arrow[dr, "\psi"] & \\ x \arrow[ur, "\phi"] \arrow[rr, "\theta"'] & & z \end{tikzcd}$$

together with a homotopy $\theta \simeq \psi\phi$ which witnesses the "commutativity" of the diagram (in higher category, commutativity of a diagram implies there are homotopies present as required to witness the composites). Simplices of higher dimension can be thought of similarly as verifying the commutivity of higher dimensional diagrams between higher level morphisms, which we will explore later as an aside from the main text.

However, this discussion has a massive issue: there is in general no way to compose two adjacent morphisms. However, inspecting the proof of proposition 1.1.2.5 gives us a property which makes this possible. A composable pair (g, f) determines a horn $\Lambda^2_1 \to K$, and the existence of their composite determines a filler $\sigma : \Delta^2 \to K$. We simply require that fillers for *inner* horns always exist.

The next problem is that of uniqueness. It is in fact fairly unnatural to require that fillers be unique. For example, recall the definition of composites of paths in a topological space, where we compress their domain along $I/2$ instead of I. We could just as easily compress one over $I/3$ and one over $2I/3$, and since the two choices are homotopic, it doesn't matter which we choose. Since they are homotopic, they should both be valid choices for the composite of these homotopies, but each corresponds to a different filler for the relevant horn. With all of these considerations, we propose the following definition.

Definition 1.1.2.8 (Quasi-category)**.** An ∞-category (in particular, a quasi-category) is a simplicial set K which admits fillers for all inner horns. More precisely, for any $0 < i < n$, for all diagrams of solid arrows

$$\begin{tikzcd}
\Lambda^n_i \arrow[r] \arrow[d, hook] & K \\
\Delta^n \arrow[ur, dotted] &
\end{tikzcd}$$

there exists at least one (not necessarily unique) dotted arrow as indicated, rendering the diagram commutative.

This definition was first proposed by Boardman and Vogt in **??**, motivated by proposition 1.1.2.5. They called them *weak Kan complexes*, and later André Joyal called them *quasi-categories*. Lurie calls them ∞-categories, although Joyal's terminology is more commonly used when referring to the specific underlying model.

Example 1.1.2.9. Any Kan complex as defined in definition 1.1.2.1 is an ∞-category. In particular, in view of remark 1.1.2.2, the singular complex $\operatorname{Sing} X$ of any topological space X is an ∞-category.

Example 1.1.2.10. By proposition 1.1.2.5, the nerve of any category is an ∞-category. We will oftentimes abuse terminology and regard 1-categories $\mathcal{C}$ as ∞-categories by identifying them with their nerve $N(\mathcal{C})$. By way of this identification, we may view 1-category theory as a special case of the theory of ∞-categories.

Notation 1.1.2.11. We will use two different notations for simplicial sets. When viewing a simplicial set K as an ∞-category, we will generally denote it with a calligraphic letter ($\mathcal{C}, \mathcal{D}, \mathcal{E} \ldots$), and otherwise we denote them by capital roman letters ($K, X, Y \ldots$). We may also employ the latter notation when comparing different models for ∞-categories, to avoid confusion.

Before we continue with the text, I want to make a couple remarks that will be useful for later.

Proposition 1.1.2.12. *Let K be a simplicial set. Specifying a horn $\Lambda^n_i \to K$ amounts to specifying $(n-1)$-simplices τ_j for all $0 \leq j \neq i \leq n$ which satisfy the compatibiltiy condition $d_k(\tau_j) = d_{j-1}(\tau_k)$ for all $k < j$. In particular, the horn is defined by $d^j \mapsto \tau_j$.*

Proof. Recall that unions are pushouts in $\mathrm{s}\mathcal{S}\mathrm{et}$, meaning morphisms $\Lambda^n_i \to K$ are precisely pushout diagrams

$$\begin{tikzcd}
\bigcap\limits_{j \in [n], j \neq i} \partial_j \Delta^n \arrow[r, hook] \arrow[d, hook] & \partial_j \Delta^n \arrow[d, hook] \arrow[ddr, bend left] & \\
\partial_k \Delta^n \arrow[r, hook] \arrow[drr, bend right] & \bigcup\limits_{j \in [n], j \neq i} \partial_j \Delta^n = \Lambda^n_i \arrow[dr, dashed] & \\
& & K
\end{tikzcd}$$

where the pushout is over all $j \in [n] \setminus \{i\}$, but it is hard to draw this many arrows so we only draw two. The case for $n = 1$ is trivial, so we assume $n \geq 2$.

Recall from the proof of proposition 1.1.2.5 that there is an isomorphism of simplicial sets $\partial_j \Delta^n \cong \Delta^{n-1}$ by mapping $\delta^j \alpha \mapsto \alpha$, which is an isomorphism by the epi-mono factorization in $\mathbf{\Delta}$ ([ML98, Tal25]). Note that the map $\delta^j \mapsto \tau_j$ decomposes as

$$\begin{tikzcd} \partial_j \Delta^n \arrow[r, "\cong", "d^j\alpha \mapsto \alpha"'] \arrow[rr, bend left, "d^j \mapsto \tau_j"] & \Delta^{n-1} \arrow[r, "1_{[n-1]} \mapsto \tau_j"'] & K, \end{tikzcd}$$

where we recall that $1_{[n-1]} \mapsto \tau_j$ is a map of simplicial sets by the Yoneda lemma, as noted in the proof of proposition 1.1.2.5. Thus, $d^j \mapsto \tau_j$ decomposes as a morphism of simplicial sets, and thus is a morphism of simplicial sets for each j. Thus by universality of pushout, and by levelwise computation of universals in functor categories ([ML98]), the claimed morphism $\sigma : \Lambda^n_i \to K$ exists precisely when the above pushout diagram commutes.

First, we prove the easy direction. Let $k < j$, and suppose the diagram commutes, meaning there is indeed a morphism $\sigma : \Lambda^n_i \to K$ with $d^j \mapsto \tau_j$ for each $j \neq i$ in $[n]$. Since Λ^n_i is a simplicial subset of Δ^n, faces are given by precomposition. Since σ is a morphism of simplicial sets, we have the string of equalities

$$d_k(\tau_j) = d_k\sigma(d^j) = \sigma d_k(d^j) = \sigma(d^j d^k) \overset{!}{=} \sigma(d^k d^{j-1}) = \sigma d_{j-1}(d^k) = d_{j-1}\sigma(d^k) = d_{j-1}(\tau_k),$$

where the equality ! is given by the cosimplicial identity $d^j d^k = d^k d^{j-1}$ for $k < j$.

Now we prove the converse. Suppose the compatibility condition holds. We do this in three steps:

1. We show for $n \geq 2$, the intersection $\bigcap_{j \neq i} \partial_j \Delta^n$ has precisely one 0-simplex i, and all higher levels m have precisely one m-simplex $s_m s_{m-1} \dots s_0(i)$.
2. We show the maps $\partial_{j,k}\Delta^n \to K$ agree on i.
3. We show this implies the maps $\partial_{j,k}\Delta^n \to K$ agree on the full intersection.

(1) First, we show i is the unique 0-simplex. Note that the 0-simplices of $\Delta^{n-1} = [n-1]$, and the 0-simplices of each $\partial_j \Delta^n$ are thus $d^j[n-1]$. Since $j \neq i$, none of the cofaces d^j skip i, meaning $i \in \operatorname{im} d^j$ for all j, and thus i is a 0-simplex. Uniqueness follows from the fact that if $k \in [n-1]$ and $k \neq i$, then there exists $\partial_k \Delta^n$ in the union, the 0-simplices of which are $d^k[n-1]$, which excludes k by definition.

Now we show all higher m-simplices are the degenerate simplices $s_m \dots s_0(i)$ by induction. A 1-simplex σ must have its faces $d_0(\sigma), d_1(\sigma)$ be 0-simplices of the union, meaning both its faces must be i. Recall that since we are working in a simplicial subset of Δ^n, faces are precomposition with cofaces, so this implies

$$0 \mapsto 0 \mapsto \sigma(0) = 0 \mapsto 1 \mapsto \sigma(1) = i,$$

and thus σ is necessarily the map $0, 1 \mapsto i$. Thus there is a unique 1-simplex. This can be viewed as the degenerate simplex

$$s_0(i) = 0, 1 \mapsto 0 \mapsto i,$$

giving the required form. Now assume the inductive hypothesis for m. Since $s_m \dots s_0(i)$ is the unique m-simplex, all faces of an $(m+1)$-simplex σ must be $s_m \dots s_0(i)$. By the same logic as above, we immediately see that σ is the map $[m+1] \mapsto i$, and thus admits the representation $s_{m+1} \dots s_0(i)$, concluding this section of the proof.

(2) Let $\sigma_j : \partial_j \Delta^n \to K$ and $\sigma_k : \partial_k \Delta^n \to K$ be the maps in question. Since n is finite, it suffices to prove the statement for the case $k = j+1$, along with the additional case $k = i+1$ and $j = i-1$.

1. ($i < j$) Since $i < j$, precomposing by d_j and d_{j+1} is the identity on $i : [0] \to [n-1]$, so undoing this precomposition gives us i as an $(n-1)$ simplex in Δ^{n-1}. In particular, it suffices to consider the maps $\sigma'_j : \Delta^{n-1} \to K$ and $\sigma'_{j+1} : \Delta^{n-1} \to K$ given by inverting the isomorphism $\partial_j \Delta^n \cong \Delta^{n-1}$. The ith vertex of the simplices $\tau_j = \sigma'_j(1_{n-1})$ and $\sigma'_{j+1}(1_{n-1})$ are the respective images of i in K. The compatibility condition gives us $d_j(\tau_j) = d_j(\tau_{j+1})$. Note that
$$d_j(\tau_j) = d_j \sigma'_j(1) = \sigma'_j(d^j).$$
Since $i < j$, recall that precomposing by d^j is an identity on the ith vertex, and by the same logic since $i < j+1$, we must have that the jth face preserves the ith vertex of τ_{j+1}. However, since $d_j(\tau_j) = d_j(\tau_{j+1})$ is a strict equality, they must have the *same* ith vertex. Thus, they agree on i,

2. ($j < i-1$) The logic for this part is the same as the above part, except d_j and d_{j+1} are no longer the identity on i. Since $j, j+1 < i-1$, they both act the same way on it, mapping $i-1$ to i. The proof is concluded from this point in the same way as above.

3. ($j = i-1$) We must now show σ_{i-1} and σ_{i+1} agree on i. Since $i-1 < i < i+1$, we have that d_{i-1} maps $i-1 \mapsto i$, and d_{i+1} maps $i \mapsto i$. Thus, we must show the $(i-1)$th vertex of τ_{i-1} is the ith vertex of τ_{i+1}, by the same logic as above. The coherence condition gives us that $d_{i-1}(\tau_{i+1}) = d_i(\tau_{i-1})$. Since $i-1 < i$, the $(i-1)$th vertex of τ_{i-1} is preserved. Since $i-1 < i+1$, the ith vertex of τ_{i+1} corresponds to the $(i-1)$th vertex of $d_{i-1}(\tau_{i+1})$, since we drop the original $(i-1)$th vertex, and the other vertices shift down. The strict equality concludes the proof.

(3) Keeping notation as before, we clearly have by inducting on the morphism condition for simplicial sets that for each m-simplex α in the intersection,
$$\sigma_j(\alpha) = \sigma_j s_m \dots s_0(i)) = s_m \dots s_0 \sigma_j(i) = s_m \dots s_0 \sigma_k(i) = \sigma_k s_m \dots s_0(i) = \sigma_k(\alpha).$$
■

Corollary 1.1.2.13. *Let K be a simplicial set. A filler for a horn $\sigma_0 : \Lambda^n_i \to K$ is an n-simplex $\tau \in K_n$ such that the morphism of simplicial sets $\sigma : 1_{[n]} \mapsto \tau$ agrees with the horn Λ^n_i in the sense that for each $j \neq i$ and each face $d^j : [n-1] \to [n]$,*
$$\sigma_0(d^j) = \sigma(d^j).$$

Proof. By the Yoneda lemma, morphisms $\sigma : \Delta^n \to K$ correspond to a choice of an n-simplex $\tau \in K_n$ and a map $1_{[n]} \mapsto \tau$. The restriction $\sigma|\Lambda^n_i$ gives a horn in K, and by proposition 1.1.2.12 this amounts to a choice τ_j for each $j \neq i$ satisfying the compatibiltiy condition, and mapping $d^j \mapsto \tau_j$. By proposition 1.1.2.12, this horn is thus the given horn σ_0 precisely when $\sigma|\Lambda^n_i$ and σ_0 agree on faces, and thus σ and σ_0 agree on ($\neq i$) faces. ■

1.1.3 Equivalences of Topological Categories

We return for a short time to our simpler notion of topological categories, in order to get a feel for what an ∞-categorical notion of an *equivalence of categories* should be. Not all Kan complexes are isomorphic to singular sets, and similarly not all CW complexes are homeomorphic to geometric realizations of simplicial sets. However, these statements *are all true* if instead of working with isomorphisms, we work with homotopy equivalences. We want to define a *functor* $F : \mathcal{C} \to \mathcal{D}$ which can be viewed as an equivalence in the same manner. Of course, the naive way to do this is to require that F induce isomorphisms of mapping spaces, meaning it is an equivalence of topological categories.

Definition 1.1.3.1. A functor $F : \mathcal{C} \to \mathcal{D}$ between topological categories is a *strong equivalence* if it is an equivalence in the sense of enriched category theory, meaning the induced continuous maps $F : \mathrm{Map}_{\mathcal{C}}(x, y) \to \mathrm{Map}_{\mathcal{D}}(F(x), F(y))$ are isomorphisms, and each $d \in \mathcal{D}$ is isomorphic to some $F(c)$ with $c \in \mathcal{C}$.

In classical category theory, this would be a perfectly fine definition, but for our cases it is still too strong. In particular, the induced morphisms between topological spaces are *isomorphisms*, not homotopy equivalences, which we would expect them to be. The obvious next idea is to recall some model category theory. Quillen equivalent model categories are generally considered to have the same homotopy theory, but the Quillen functors need not actually be equivalences; they need only lower to equivalences of the homotopy categories. Since geometric realization and the singular complex form a Quillen equivalence bewteen $\mathcal{S}\mathrm{et}$ and $\mathcal{T}\mathrm{op}$, this line of thought is very promising. However, this requires a definition of a homotopy category. To this end, we propose definition 1.1.3.2, which seems quite reasonable given that we are working with *weak* homotopy equivalences. Recall that $\pi_0 : \mathcal{T}\mathrm{op} \to \mathcal{S}\mathrm{et}$ maps topological spaces to their set of path components.

Definition 1.1.3.2. Let $\mathcal{C}$ be a topological category. The *homotopy category* $\mathrm{h}\mathcal{C}$ of $\mathcal{C}$ is defined by

- $\mathrm{Obj}(\mathrm{h}\mathcal{C}) = \mathrm{Obj}(\mathcal{C})$;
- $\mathrm{h}\mathcal{C}(x, y) = \pi_0 \mathrm{Map}_{\mathcal{C}}(x, y)$;

and composition is inherited by functoriality of π_0. In particular, $\mathrm{h}\mathcal{C} = \mathrm{im}\, \pi_0$.

Example 1.1.3.3. Let $\mathcal{C}$ be the category of CW complexes, with mapping spaces given by the k-ification of the compact-open topology. The homotopy category thereof is quite important, is denoted by $\mathcal{H}$, and is called the *homotopy category of spaces*.

Using $\mathcal{H}$, we can rewrite definition 1.1.3.2 in a useful way. Recall:

Definition 1.1.3.4. A map $f : X \to Y$ of topological spaces is a *weak homotopy equivalence* if it induces a bijection $\pi_0(f) : \pi_0(X) \to \pi_0(Y)$ on path components, and isomorphisms on all homotopy groups

$$\pi_i(f) : \pi_i(X, x) \to \pi_i(Y, f(x)).$$

Classical homotopy theory tells us for each $X \in \mathcal{CG}$, there is a CW complex X' and a weak homotopy equivalence $\varphi : X' \to X$. This space X' is well-defined up to canonical weak homotopy equivalence, meaning there is a functor $\theta : \mathcal{CG} \to \mathcal{H}$ given by $X \mapsto [X] \ni X'$. This is precisely the localization functor, meaning it is universal with respect to inverting weak homotopy equivalences. Recall that lax monoidal functors $F : \mathcal{C} \to \mathcal{C}'$ induce 2-functors $F : \mathcal{C}\text{-}\mathcal{C}\mathrm{at} \to \mathcal{C}'\text{-}\mathcal{C}\mathrm{at}$, and since θ fairly clearly preserves products, θ induces a 2-functor $\mathcal{C}\mathrm{at}_{\mathrm{top}} \to \mathcal{H}\text{-}\mathcal{C}\mathrm{at}$. In particular, topological categories $\mathcal{C}$ are mapped to $\mathcal{H}$-enriched categories. It seems reasonable, since θ is localization, to define the image of $\mathcal{C}$ as the *homotopy category* $\mathrm{h}\mathcal{C}$ of $\mathcal{C}$. Recalling how the functor $\mathcal{C}\mathrm{at}_{\mathrm{top}} \to \mathcal{H}\text{-}\mathcal{C}\mathrm{at}$ works, this gives

1. $\mathrm{Obj}(\mathrm{h}\mathcal{C}) = \mathrm{Obj}(\mathcal{C})$;

2. For $x, y \in \mathcal{C}$, we have
$$\mathrm{Map}_{\mathrm{h}\mathcal{C}}(x, y) = [\mathrm{Map}_{\mathcal{C}}(x, y)],$$
where we continue to use the equivalence class $[-]$ notation to denote a choice of a weakly homotopic equivalent CW complex.
3. Composition is given by functoriality of $\theta : \mathcal{CG} \to \mathcal{H}$.

Note here that θ maps morphisms to their (regular) homotopy classes. For a refresher on enriched category theory, we recommed chapter 3 of [Rie14], but for a more general account of the theory see [Kel05].

Intuition 1.1.3.5. Departing slightly from the book here to improve my intuition. θ is simply localization at the class of weak homotopy equivalences, which is the class of weak equivalences in $\mathcal{T}\mathrm{op}$. By Whitehead's theorem, we may choose $\mathcal{H}$ to have CW complexes as objects up to equivalence of categories. Of course, between cofibrant and fibrant objects, this agrees with the regular notion of a homotopy equivalence. In particular, each $\pi_n(X, x)$ identifies homotopic maps, making homotopy equivalence a stronger condition than weak homotopy equivalence.

Remark 1.1.3.6. The equivalence of our definition of $\mathrm{h}\mathcal{C}$ by θ and by definition 1.1.3.2 follows by noting that there is a canonical isomorphism $\pi_0(X) \cong \mathrm{Map}_{\mathcal{H}}(*, [X])$, recovering $\mathrm{h}\mathcal{C}$ as the underlying category of the $\mathcal{H}$-enriched version.

Taking our cue from model category theory, we now think to define an equivalence as a functor that descends to an equivalence of categories $\mathrm{h}\mathcal{C} \simeq \mathrm{h}\mathcal{D}$.

Definition 1.1.3.7. Let $F : \mathcal{C} \to \mathcal{D}$ be a functor between topological categories. We say F is a *weak equivalence* if the induced functor $\mathrm{h}F : \mathrm{h}\mathcal{C} \to \mathrm{h}\mathcal{D}$ is an equivalence of $\mathcal{H}$-categories. Equivalently, F is an equivalence if and only if

1. for each $x, y \in \mathcal{C}$, the map $\mathrm{Map}_{\mathcal{C}}(x, y) \to \mathrm{Map}_{\mathcal{D}}(F(x), F(y))$ is a weak homotopy equivalence;
2. Each $d \in \mathcal{D}$ is isomorphic to $F(c)$ for some $c \in \mathcal{C}$.

Here we recall that an equivalence of $\mathcal{H}$-categories must induce isomorphisms on hom-objects, and since θ inverts weak homotopy equivalences, the maps can equivalently be weak homotopy equivalences in $\mathcal{CG}$. Weak equivalences are the correct notion of equivalences in the context of higher category theory. We will thus refer to them simply as *equivalences* as we continue. Of course, two topological categories are *equivalent* if an equivalence between them exists. However, before we continue, note that the homotopy category of a topological category $\mathcal{C}$ does *not* determine $\mathcal{C}$ up to equivalence.

1.1.4 Simplicial Categories

Now that we have a good notion of equivalences of topological categories, we want to show the theories of quasi-categories and topological categories are equivalent. It will be extremely beneficial to introduce the theory of *simplicial categories*, which is not only related to both approaches, but will also motivate definitions similar to those of section 1.1.3 for quasi-categories.

Definition 1.1.4.1. A *simplicial category* is a $\mathrm{s}\mathcal{S}\mathrm{et}$-category; that is, it is a category enriched over the category of simplicial sets. We denote the category of simplicial categories by $\mathrm{s}\mathcal{C}\mathrm{at}$.

Remark 1.1.4.2. Every simplicial category $\mathcal{C}$ forms a simplicial object $\mathcal{C}_\bullet$ in $\mathcal{C}\mathrm{at}$ by taking $\mathrm{Obj}(\mathcal{C}_n) = \mathrm{Obj}(\mathcal{C})$, and $\mathcal{C}_n(x, y) = \mathcal{C}(x, y)_n$. Faces and degeneracies are defined by the action of the faces and degeneracies of $\mathcal{C}$ within its hom-objects. Conversely, simplicial objects $\mathcal{C}_\bullet$ in $\mathcal{C}\mathrm{at}$ give rise to simpliical categories $\mathcal{C}$ *precisely when* $\mathrm{Obj}(\mathcal{C}_n) = \mathrm{Obj}(\mathcal{C}_m)$ for all n, m. We say this means the *underlying simplicial set* of objects is constant. Also note that every category is a simplicial category by considering the discrete simplicial set on each hom-set.

Of course, the Quillen adjunction $|-| \dashv \mathrm{Sing} : \mathrm{s}\mathcal{S}\mathrm{et} \to \mathcal{C}\mathcal{G}$ makes the relationship between topological and simplicial categories simple to describe. Both of these functors preserve finite products, and thus are lax monoidal functors. Thus, given a simplicial category $\mathcal{C}$, there is a topological category $|\mathcal{C}|$ given by

- $\mathrm{Obj}(\mathcal{C}) = \mathrm{Obj}(|\mathcal{C}|)$;
- For all $x, y \in \mathcal{C}$, $\mathrm{Map}_{|\mathcal{C}|}(x, y) = |\mathrm{Map}_{\mathcal{C}}(x, y)|$.

We also obtain for each topological category $\mathcal{C}$ a simplicial category $\mathrm{Sing}\,\mathcal{C}$ given in the same way by

- $\mathrm{Obj}(\mathcal{C}) = \mathrm{Obj}(\mathrm{Sing}\,\mathcal{C})$;
- For all $x, y \in \mathcal{C}$, $\mathrm{Map}_{\mathrm{Sing}\,\mathcal{C}}(x, y) = \mathrm{Sing}(\mathrm{Map}_{\mathcal{C}}(x, y))$.

In particular, $|-| \dashv \mathrm{Sing} : \mathrm{s}\mathcal{S}\mathrm{et} \to \mathcal{C}\mathcal{G}$ raises to an adjunction $|-| \dashv \mathrm{Sing} : \mathrm{s}\mathcal{C}\mathrm{at} \to \mathcal{C}\mathrm{at}_{\mathrm{top}}$. To show the theories of topological and simplicial categories are equivalent, since we have modeled them homotopy-theoretically, we once again turn to homotopy theory. We previously defined weak equivalences of topological categories, so now we would like to define them between simplicial categories. We recall that weak equivalences between simplicial sets are maps $f : X \to Y$ which induce weak homotopy equivalences $|f| : |X| \to |Y|$. Inverting these gives a homotopy category which we also denote $\mathcal{H}$, since the Quillen equivalence $|-| \dashv \mathrm{Sing}$ defines an equivalence of categories between the homotopy categories for CW complexes and for simplicial sets.

We can once again define $\mathrm{h}\mathcal{C}$ for simplicial categories by applying the functor $\mathrm{s}\mathcal{S}\mathrm{et} \to \mathcal{H}$ and considering the induced $\mathcal{H}$-category. There are canonical isomorphisms $\mathrm{h}\mathcal{C} \cong \mathrm{h}\,|\mathcal{C}|$ and $\mathrm{h}\mathcal{D} \cong \mathrm{h}\,\mathrm{Sing}\,\mathcal{D}$, meaning that our choice of notation offers little risk of confusion. It is at this point fairly obvious that topological categories and simplicial categories are equivalent, but we may finalize this fact by introducing the following definition.

Definition 1.1.4.3. A functor $F : \mathcal{C} \to \mathcal{D}$ between simplicial categories is an *equivalence* if $\mathrm{h}F : \mathrm{h}\mathcal{C} \to \mathrm{h}\mathcal{D}$ is an equivalence of $\mathcal{H}$-categories.

In particular, since $\mathrm{h}\mathcal{C} \cong \mathrm{h}\,|\mathcal{C}|$, a functor $F : \mathcal{C} \to \mathcal{D}$ is an equivalence if and only if the geometric realization $|F| : |\mathcal{C}| \to |\mathcal{D}|$ is an equivalence. Even better, we note by these remarks that in the adjunction $|-| \dashv \mathrm{Sing} : \mathrm{s}\mathcal{C}\mathrm{at} \to \mathcal{C}\mathrm{at}_{\mathrm{top}}$, the unit and counit maps

$$\mathcal{C} \to \mathrm{Sing}\,|\mathcal{C}|\,, \qquad |\mathrm{Sing}\,\mathcal{C}| \to \mathcal{C}$$

induce *isomorphisms of homotopy categories*. Not only is this exactly what we expect a Quillen equivalence to look like (without defining a whole model structure), but we also note that we may freely replace topological categories with simplicial categories up to equivalence, giving us our equivalence of models for ∞-categories.

1.1.5 Comparing ∞-Categories with Simplicial Categories

In section 1.1.5, we showed simplicial categories and topological categories were equivalent as models for ∞-categories. Showing this amounted to showing that there is an adjunction which is similar to a Quillen equivalence in that it preserves (weak) equivalences, and that the unit and counit are natural (weak) equivalences, allowing us to replace topological categories $\mathcal{C}$ with simplicial categories $\operatorname{Sing}\mathcal{C}$, and simplicial categories $\mathcal{C}$ with topological categories $|\mathcal{C}|$, up to equivalence. Our proof that simplicial categories are equivalent to quasi-categories should be similar. I also remark here that these equivalences *are indeed Quillen equivalences*, but since this first section of the book is more of an overview, we're glossing over many of the details, including (co)fibrations. Our proof will involve the *simplicial nerve functor* $N : \mathrm{s}\mathcal{C}\mathrm{at} \to \mathrm{s}\mathcal{S}\mathrm{et}$, and the *Joyal rigification functor* $\mathfrak{C} : \mathrm{s}\mathcal{S}\mathrm{et} \to \mathrm{s}\mathcal{C}\mathrm{at}$. Credit to the nLab ([nLa25]) for the name of $\mathfrak{C}$.

Recall that the nerve of a category $\mathcal{C}$ has n-simplices $N(\mathcal{C})_n$ given by chains of n composable morphisms. This is equivalently viewed as a functor $[n] \to \mathcal{C}$, since the datum of such a functor amounts to a choice of $n+1$ objects, and n composable morphisms between them. Also recalling that n-simplices are precisely morphisms $\Delta^n \to N(\mathcal{C})$, we can characterize n-simplices via the natural isomorphism

$$\mathrm{s}\mathcal{S}\mathrm{et}(\Delta^n, N(\mathcal{C})) \cong \mathcal{C}\mathrm{at}([n], \mathcal{C}) = N(\mathcal{C})_n.$$

We would like to promote this to a relationship between simplicial categories. Rather than waste time trying to figure out how to do this, we are just going to note that we end up with the wrong idea. The promotion we get does not involve the simplicial structure of $\mathcal{C}$, and instead it's more suitable to replace $[n]$ with a different category $\mathfrak{C}[\Delta^n]$.

Definition 1.1.5.1. Let J be a finite nonempty linearly ordered set. The simplicial category $\mathfrak{C}[\Delta^J]$ is defined by

1. The objects of $\mathfrak{C}[\Delta^J]$ are the elements of J.
2. For $i, j \in J$, mapping spaces are given by

$$\mathrm{Map}_{\mathfrak{C}[\Delta^J]}(i,j) = \begin{cases} \varnothing, & j < i \\ N(P_{i,j}), & i \leq j \end{cases},$$

   where P_{ij} is the poset

$$P_{ij} = \{I \subseteq J \mid i, j \in I \wedge (\forall k \in I)(i \leq k \leq j)\}$$

   ordered by inclusion. Note that elements I have $\min I = i$ and $\max I = j$.
3. If a mapping space is empty then a composition morphism to/from it is the empty map, so let $i_0 \leq \ldots \leq i_n$. Composition morphisms

$$\mathrm{Map}_{\mathfrak{C}[\Delta^j]}(i_0, i_1) \times \cdots \times \mathrm{Map}_{\mathfrak{C}[\Delta^J]}(i_{n-1}, i_n) \to \mathrm{Map}_{\mathfrak{C}[\Delta^J]}(i_0, i_n)$$

   are given by taking the nerve functor on the morphism of poset categories

$$P_{i_0,i_1} \times \cdots \times P_{i_{n-1},i_n} \to P_{i_0,i_n},$$

$$(I_1, \ldots, I_n) \mapsto \bigcup_{j \in [n]} I_j.$$

Lurie offers a short description of the topological category $|\mathfrak{C}[\Delta^n]|$ to provide some intuition. The objects of $|\mathfrak{C}[\Delta^n]|$ are the objects of $\mathfrak{C}[\Delta^n]$ are the objects of $[n]$. The space $\mathrm{Map}_{|\mathfrak{C}[\Delta^n]|}(i,j)$ is homeomorphic to a cube when $i \leq j$, and empty otherwise. It may be identified with the set of functions $p : \{k \in [n] \mid i \leq k \leq j\} \to I$ such that $p(i) = p(j) = 1$, and composition is given by concatenation of functions.

Now we try to understand the simplicial category $\mathfrak{C}[\Delta^n]$ itself, and how it relates to $[n]$. Once again, $\mathfrak{C}[\Delta^n]$ and $[n]$ have the same objects. In $[n]$, there are unique morphisms $i \to j$ for $i \leq j$, and composites are defined by virtue of this uniqueness. We denote by q_{ij} the morphism $i \to j$, and note that $q_{jk} \circ q_{ij} = q_{ik}$ for $i \leq j \leq k$.

In $\mathfrak{C}[\Delta^n]$, we have vertices $p_{ij} \in \mathrm{Map}_{\mathfrak{C}[\Delta^n]}(i,j)$ for each $i \leq j$ given by $\{i,j\} \in P_{ij}$. Of course taking the view expressed above, $p_{jk} \circ p_{ij} = \{i,j,k\} \neq p_{ik}$ for $i \leq j \leq k$, unless we have a degenerate case with either $i = j$ or $j = k$. Instead, all compositions of the form

$$p_{i_{n-1}i)n} \circ \dots \circ p_{i_0 i_1}$$

with $i_n < i_m$ for $n < m$ constitute the vertices of the cube $\mathrm{Map}_{\mathfrak{C}[\Delta^n]}(i,j)$. Since the cube is contractible (i.e. the identity is null-homotopic), it is in particular weakly contractible, and thus although the vertices of the cube are not necessarily the same, they are all canonically homotopic to each other by the Quillen equivalence $|-| \dashv \mathrm{Sing}$.

Indeed, there is a unique functor $\mathfrak{C}[\Delta^n] \to [n]$ which is the identity on objects, and identifies homotopic compositions. In summary, $\mathfrak{C}[\Delta^\bullet]$ is a *thickened* version of $[n]$, where composition is no longer strictly associative, and rather is associative up to (coherent) homotopy. Lurie notes that this is stated more precisely by saying $\mathfrak{C}[\Delta^\bullet]$ is a cofibrant replacement for $[\bullet]$ in $\mathrm{s}\mathcal{C}\mathrm{at}$. I don't 100% see the connection, but that's fine. We are more interested in showing $J \mapsto \mathfrak{C}[\Delta^J]$ is functorial $\mathcal{L}\mathrm{in} \to \mathrm{s}\mathcal{C}\mathrm{at}$.

Definition 1.1.5.2. Let $f : J \to J'$ be a morphism of posets. The simplicial functor $\mathfrak{C}[f] : \mathfrak{C}[\Delta^J] \to \mathfrak{C}[\Delta^{J'}]$ is given by

- On objects $i \in \mathfrak{C}[\Delta^J]$ (and thus $i \in J$), we define $\mathfrak{C}[f](i) = f(i)$.
- If $i \leq j$ in J, then the map $\mathrm{Map}_{\mathfrak{C}[\Delta^J]}(i,j) \to \mathrm{Map}_{\mathfrak{C}[\Delta^{J'}]}(f(i), f(j))$ induced by f is the nerve of

$$P_{i,j} \to P_{f(i),f(j)},$$
$$I \mapsto f(I).$$

Claim 1.1.5.3. $\mathfrak{C}[f]$ *is functorial.*

Proof. By functoriality of the nerve, it suffices to show the given map of poset categories is functorial. Indeed, if f is the identity 1, then the morphism function is $I \mapsto I$, preserving the identity. Composition is preserved since $f(I_1) \cup f(I_2) = f(I_1 \cup I_2)$ is a well-known fact in set theory.

I assumed this would suffice as a proof but was not fully confident, so I wrote the argument out again in more rigor. I include it here because I can. We very obviously have that identities are preserved, so it suffices to show that $\mathfrak{C}[f]_{i,j} : I \mapsto f(I)$ (where we abuse notation mildly since we are not working in the nerve) preserves composition $(I_1, I_2) \mapsto I_1 \cup I_2$. In enriched category theory, this amounts to the commutativity of the diagram

$$\begin{tikzcd}
P_{j,k} \times P_{i,j} \arrow[r, "\circ"] \arrow[d, "\mathfrak{C}[f]_{j,k} \times \mathfrak{C}[f]_{i,j}"'] & P_{i,k} \arrow[d, "\mathfrak{C}[f]_{i,k}"] \\
P_{f(j),f(k)} \times P_{f(i),f(j)} \arrow[r, "\circ"'] & P_{f(i),f(k)}
\end{tikzcd}$$

and tracing a pair (I_1, I_2) around the square shows that this amounts to the equality noted above. ■

Remark 1.1.5.4. Note that $p_{i,j} = \{i,j\}$ is mapped by $\mathfrak{C}[f]$ to $p_{f(i),f(j)} = \{f(i), f(j)\}$.

Since $\mathfrak{C} : \mathcal{L}\mathrm{in} \to \mathrm{s}\mathcal{C}\mathrm{at}$ is a functor, restricting along the simplex category gives a cosimplicial object $\mathfrak{C} : \mathbf{\Delta} \to \mathrm{s}\mathcal{C}\mathrm{at}$ with $[n] \mapsto \mathfrak{C}[\Delta^n]$.

Definition 1.1.5.5. Let $\mathcal{C}$ be a simplicial category. Define the *simplicial nerve* $N(\mathcal{C})$ to be the simplicial set with n-simplices given by

$$N(\mathcal{C})_n := \mathrm{s}\mathcal{C}\mathrm{at}(\mathfrak{C}[\Delta^n], \mathcal{C}).$$

If $\mathcal{C}$ is a topological category, define the *topological nerve* $N(\mathcal{C})$ as the simplicial nerve of $\mathrm{Sing}\,\mathcal{C}$.

Definition 1.1.5.5 realizes our goal of defining a more suitable nerve for simplicial categories, and using the Quillen equivalence of simplicial and topological categories, we also obtain a topological nerve. Note that by the Yoneda lemma, there is an isomorphism

$$\mathrm{s}\mathcal{S}\mathrm{et}(\Delta^n, N(\mathcal{C})) \cong \mathrm{s}\mathcal{C}\mathrm{at}(\mathfrak{C}[\Delta^n], \mathcal{C}).$$

It would be nice to show the simplicial nerve is functorial, since we would then have an adjunction. Before doing this however, we discuss terminology and present some intuition.

Terminology 1.1.5.6. If $\mathcal{C}$ is a simplicial or topological category, we often refer to the simplicial or topological nerve of $\mathcal{C}$ simply as the *nerve* of $\mathcal{C}$. Note that this does not necessarily coencide with the ordinary nerve of the underlying category, so we adopt the convention of writing $N(\mathcal{C})$ to mean the *simplicial* or *topological* nerve of $\mathcal{C}$ if $\mathcal{C}$ is simplicial topological, and if we need to consider the regular nerve of the underlying category, we will always specify that.

We continue by considering the nerve of a topological category $\mathcal{C}$, which is the nerve of the simplicial category $\mathrm{Sing}\,\mathcal{C}$. 0-simplices are morphisms $\mathfrak{C}[\Delta^0] \to \mathrm{Sing}\,\mathcal{C}$. Recall that $\mathfrak{C}[\Delta^0]$ takes as objects $[0] = *$, and as its mapping space $P_{0,0} = \{0\} = *$, implying that $N(P_{0,0})$ is the discrete simplicial set on the singleton, which is isomorphic to Δ^0. Thus, a morphism $\mathfrak{C}[\Delta^0] \to \mathrm{Sing}\,\mathcal{C}$ is an object x of $\mathrm{Sing}\,\mathcal{C}$ (equivalently an object of $\mathcal{C}$) and a 0-simplex of the singular set of the mapping space $\mathrm{Sing}\,\mathrm{Map}_{\mathcal{C}}(x, x)$. Since the simplicial functor must preserve identity morphisms, and since the nerve $N(P_{0,0})$ is the identity in $\mathfrak{C}[\Delta^0]$, we have that the 0-simplex selected must be the 0-simplex in $\mathrm{Sing}\,\mathrm{Map}_{\mathcal{C}}(x, x)$ corresponding to the identity in $\mathrm{Map}_{\mathcal{C}}(x, x)$. Thus, *0-simplices of the topological nerve are precisely object of* $\mathcal{C}$.

Continuing, we now unpack 1-simplices of $N(\mathcal{C})$. A morphism $\mathfrak{C}[\Delta^1] \to \mathrm{Sing}\,\mathcal{C}$ is a choice of two objects x, y in $\mathrm{Sing}\,\mathcal{C}$, which are mapped to by 0 and 1, respectively; and a 1-simplex of $\mathrm{Sing}\,\mathrm{Map}_{\mathcal{C}}(x, y)$, since the only non-identity mapping space is $N(P_{0,1})$. Since morphisms $x \to y$ in the underlying category of $\mathcal{C}$ are points in $\mathrm{Map}_{\mathcal{C}}(x, y)$, the datum of a 1-simplex is a morphism $x \to y$ in the underlying category.

We now consider the 2-simplices of $N(\mathcal{C})$. Morphisms $\mathfrak{C}[\Delta^2] \to \mathrm{Sing}\,\mathcal{C}$ identify three objects $0 \mapsto x$, $1 \mapsto y$, and $2 \mapsto z$, and a 2-simplex $\tau \in \mathrm{Sing}\,\mathrm{Map}_{\mathcal{C}}(x, z)$. The simplicial identities guarentee that the boundary of this 2-simplex constitutes the three 1-simplices the functor decides, one in each of $\mathrm{Sing}\,\mathrm{Map}_{\mathcal{C}}(x, y)$, $\mathrm{Sing}\,\mathrm{Map}_{\mathcal{C}}(x, z)$, and $\mathrm{Sing}\,\mathrm{Map}_{\mathcal{C}}(y, z)$. Write $f_{x,y}$, $f_{x,z}$, and $f_{y,z}$ for these 1-simplices, respectively. The 2-simplex then corresponds to a path from $f_{x,z}$ to $f_{y,z} \circ f_{x,y}$ in the mapping space $\mathrm{Map}_{\mathcal{C}}(x, z)$.

This is enough analysis for the reader to identify the relevant analogues with our treatment of morphisms in quasi-categories. The reader is likely expecting, due to our choice of notation, that we should be able to compute $\mathfrak{C}$ on any simplicial set, not just representable simplicial sets for linearly ordered sets. Indeed, recall that by [ML98, III.7], any functor is a colimit of representable functors, so simply define $\mathfrak{C} : \mathrm{sSet} \to \mathrm{s}\mathcal{C}\mathrm{at}$ as a colimit preserving functor, whose action on Δ^n agrees with our above notation, essentially extending $\mathfrak{C}$ along the Yoneda embedding. This exists since $\mathrm{s}\mathcal{C}\mathrm{at}$ is cocomplete. By construction, $\mathfrak{C} \dashv N$ for N the simplicial nerve, viewed as a functor in the obvious way.

I would rather not compute examples of this category if it is not required. To make computation simpler, we can note that $\mathfrak{C}$ is the left Kan extension of $\mathfrak{C}[\Delta^\bullet]$ along the Yoneda embedding.

Proposition 1.1.5.7. *Let $\mathcal{C}$ be a simplicial category such that each mapping space is a Kan complex. Then its simplicial nerve $N(\mathcal{C})$ is an ∞-category.*

Proof. Let $0 < i < n$; we must show $N(\mathcal{C})$ has the right lifting property with respect to inclusions $\Lambda^n_i \subseteq \Delta^n$. By the adjunction $\mathfrak{C} \dashv N$, we may equivalently show $\mathcal{C}$ has the right extension property with respect to the simplicial functor $\mathfrak{C}[\Lambda^n_i] \to \mathfrak{C}[\Delta^n]$. Proof of the equivalence of these statements is done simply via isomorphisms of hom-sets, but the diagrammatic proof via units and counits is more illuminating. Since I have done similar proofs, I omit writing this up in my notes.

We start by considering the structure of $\mathfrak{C}[\Lambda^n_i]$. I take Lurie at his word for the following observations, to avoid needing to compute the relevant colimits.

- Objects of $\mathfrak{C}[\Lambda^n_i]$ are objects of $\mathfrak{C}[\Delta^n]$, which are objects of $[n]$.
- For $0 < j \leq k < n$, the mapping space $\mathrm{Map}_{\mathfrak{C}[\Lambda^n_i]}(j,k) = \mathrm{Map}_{\mathfrak{C}[\Delta^n]}(j,k)$. This fails when $j = 0$ or $k = n$ (or when the horn isn't an inner horn).

The lifting problem thus reduces to the action of the functor on the mapping spaces

$$\begin{tikzcd}
\mathrm{Map}_{\mathfrak{C}[\Lambda^n_i]}(0,n) \arrow[r] \arrow[d] & \mathrm{Map}_{\mathcal{C}}(F(0),F(n)) \\
\mathrm{Map}_{\mathfrak{C}[\Delta^n]}(0,n) \arrow[ur, dotted] &
\end{tikzcd}$$

where $F : \Lambda^n_i \to N(\mathcal{C})$ was the original horn. The mapping space on the right is a Kan complex by hypothesis, which concludes the proof. ■

Corollary 1.1.5.8. *Let $\mathcal{C}$ be a topological category. Then $N(\mathcal{C})$ is an ∞-category.*

We are now prepared to cite the following theorem, but we cannot yet prove it.

Theorem 1.1.5.9. *Let $\mathcal{C}$ be a topological category, and let $x, y \in \mathcal{C}$. Then the counit*

$$\left|\mathrm{Map}_{\mathfrak{C}[N(\mathrm{Sing}(\mathcal{C}))]}(x,y)\right| \to \mathrm{Map}_{\mathcal{C}}(x,y)$$

is a weak homotopy equivalence, where N is the simplicial nerve.

Note that this is the composite adjunction of $|-| \dashv \mathrm{Sing}$ and $\mathfrak{C} \dashv N$. This shows that the theory of ∞-categories is equivalent to that of topological categories or simplicial categories. Although $|\mathfrak{C}[-]| \dashv N$ is not an equivalence, they are *homotopy* inverses of each other. Making this more precise motivates the definition:

Definition 1.1.5.10. Let S be a simplicial set. The *homotopy category* $\mathrm{h}S$ is defined to be the homotopy category of the simplicial category $\mathrm{h}\mathfrak{C}[S]$, which we recall is given by the $\mathcal{H}$-category on $\mathfrak{C}[S]$ induced by lax-monoidality of the quotient functor $\gamma : \mathrm{sSet} \to \mathrm{Ho}\,\mathrm{sSet}$ (mapping spaces are given by computing γ on them).

In particular, for vertices $x, y \in S$, the mapping space of the homotopy category has $\mathrm{Map}_{\mathrm{h}S}(x, y) = [\mathrm{Map}_{\mathfrak{C}[S]}(x, y)]$ the class of path components of $\mathrm{Map}_{\mathfrak{C}[S]}(x, y)$, given of course by the singular set of the functor π_0 (classes of vertices with two vertices are equivalent if there is a 1-simplex between them). This is of course a set, and composition morphisms are induced in the expected way, rendering the homotopy category as a normal 1-category.

Definition 1.1.5.11. A map $f : S \to T$ of simplicial sets is a *categorical equivalence* if the induced map $\mathrm{h}f : \mathrm{h}S \to \mathrm{h}T$ is an equivalence of $\mathcal{H}$-categories.

Lurie notes that Joyal calls the notion given in definition 1.1.5.11 *weak categorical equivalences*. I do not know how the modern literature has evolved in this regard. However, this seems fairly standard compared to other terminology, and I would be suprised if the literature has evolved in favor of Joyal's terminology. We call these maps *categorical* equivalences to avoid conflating terminology with weak equivalences of simplicial sets.

Remark 1.1.5.12. From definition 1.1.5.11, we immediately have that $f : S \to T$ is an equivalence if and only if $\mathfrak{C}[f] : \mathfrak{C}[S] \to \mathfrak{C}[T]$ is an equivalence of simplicial categores if and only if $|\mathfrak{C}[f]| : |\mathfrak{C}[S]| \to |\mathfrak{C}[T]|$ is an equivalence of topological categories (by theorem 1.1.5.9).

This asserts that the counnit $|\mathfrak{C}[N(\mathrm{Sing}(\mathcal{C}))]| \to \mathcal{C}$ is an equivalence of topological categories, and the unit $S \to \mathrm{Sing}(N\,|\mathfrak{C}[S]|$ is a categorical equivalence of simplicial sets. Note that by theorem 1.1.5.9, every simplicial set is canonically categorically equivalent to an ∞-category, proving the equivalence of these models as theories for $(\infty, 1)$-categories.

1.2 The Language of Higher Category Theory

This section is devoted to defining many elementary 1-categorical notions in an ∞-categorical setting. Our main results are the defintions of the opposite category, mapping spaces, the homotopy category, the language of objects, morphisms, and equivalences, homotopy, (full, faithful, essentially surjective) functors, joins, slice and coslice categories, subcategories, initial and terminal objects, limits and colimits, and presentations.

1.2.1 The Opposite of an ∞-Category

Definition 1.2.1.1. Lurie's definition in [Lur09] made little sense, so I pulled from [Lur25, 003L]. Let S be a simplicial set. Define $\mathrm{Op} : \mathbf{\Delta} \to \mathbf{\Delta}$ by $[n] \mapsto [n]$ and for each $\alpha : [m] \to [n]$, we define $\mathrm{Op}(\alpha) : [m] \to [n]$ by mapping $i \mapsto n - \alpha(m - i)$. We then define $S^{\mathrm{op}} = S \circ \mathrm{Op}$.

Equivalently, and more concretely, we define $S^{\mathrm{op}}_n = S_n$, and faces and degeneracies are given by

$$(d_i : S^{\mathrm{op}}_n \to S^{\mathrm{op}}_{n-1}) = (d_{n-1} : S_n \to S_{n-1}),$$
$$(s_i : S^{\mathrm{op}}_n \to S^{\mathrm{op}}_{n+1}) = (s_{n-i} : S_n \to S_{n+1}).$$

Claim 1.2.1.2. *Let S be an ∞-category. Then so is S^{op}.*

Proof. We show for $0 < i < n$, S fills horns $\Lambda^n_i \to S$ if and only if S^{op} fills horns Λ^n_{n-i}. First, we must of course show S^{op} is a simplicial set, and since $S \circ \mathrm{Op}^{\mathrm{op}} : \mathbf{\Delta}^{\mathrm{op}} \to \mathcal{S}\mathrm{et}$, it suffices to show faces and degeneracies are indeed described as claimed in definition 1.2.1.1. For this, it suffices to show $\mathrm{Op}(d^i) = d^{n-i}$ and $\mathrm{Op}(s^i) = s^{n-i}$ in $\mathbf{\Delta}$. We prove this only for faces, since the proof for degeneracies is the same. Let $j \in [n-1]$, and note that by definition 1.2.1.1,

$$\mathrm{Op}(d^i)(j) = n - d^j(n-1-j) = \begin{cases} n-n+1+j, & n-1-j < i \\ n-n+1+j-1, & n-1-j \geq i \end{cases}.$$

Note that the top case simplifies to $j+1$ and the bottom to j. Also note that $n-1-j < i$ if and only if $n-i < j+1$ if and only if $n-i \leq j$. Similarly, $n-1-j \geq i$ if and only if $n-i \geq j+1$ if and only if $n-i > j$. This is precisely the definition of d^{n-i}. Thus, the definitions as given are in agreement, and in particular this shows Op is indeed a functor. Thus, S^{op} is a simplicial set.

We now show the horn filling condition. Let S be an ∞-category. This first involves showing $\mathrm{Op} : \mathbf{\Delta} \to \mathbf{\Delta}$ is an isomorphism. Indeed, it is it's own inverse, since

$$(\mathrm{Op} \circ \mathrm{Op})[n] = [n], \qquad (\mathrm{Op} \circ \mathrm{Op})(d^i) = \mathrm{Op}(d^{n-i}) = d^{n-(n-i)} = d^i,$$

$$(\mathrm{Op} \circ \mathrm{Op})(s^i) = \mathrm{Op}(s^{n-i}) = s^{n-(n-i)} = s^i.$$

By Yoneda, precomposition gives an isomorphism $(\mathrm{Op}^{\mathrm{op}})^* : \mathcal{C}\mathrm{at}(\mathbf{\Delta}^{\mathrm{op}}, -) \cong \mathcal{C}\mathrm{at}(\mathbf{\Delta}^{\mathrm{op}}, -)$. Temporarily passing to a higher universe, we may plug $\mathcal{S}\mathrm{et}$ into the blank, giving $(\mathrm{Op}^{\mathrm{op}})^* : \mathrm{s}\mathcal{S}\mathrm{et} \cong \mathrm{s}\mathcal{S}\mathrm{et}$. We have thus reduced the problem to showing for any inner horn $\sigma_0 : \Lambda^n_i \to S^{\mathrm{op}}$, the corresponding map $\sigma_0^{\mathrm{op}} : (\Lambda^n_i)^{\mathrm{op}} \to S$ is also a horn, and that $(\Delta^n)^{\mathrm{op}} \cong \Delta^{\mathrm{op}}$.

We first show $\Delta^n \cong (\Delta^n)^{\mathrm{op}}$. Actually I dont wanna prove this the map is precompositing by $i \mapsto m - i$ for $i \in [m]$ go do the proof urself ok cool

We note that restricting this isomorphism along a horn Λ^n_i gives $d^j \mapsto d^{n-j}$. In particular, $\Lambda^n_i \cong (\Lambda^n_{n-i})^{\mathrm{op}}$ (given by a simple inspection of elements). Since $0 < i$, we have $n - i < n$, and since $i < n$, we have $0 < n - i$. Thus, Λ^n_{n-i} is an inner horn, and admits a filler under σ_0^{op} since S is an ∞-category, completing the proof. ■

We will use the fact that $\mathrm{Op} : \mathbf{\Delta} \cong \mathbf{\Delta}$ thoughout the text to promote the principle of duality to an ∞-categorical setting.

1.2.2 Mapping Spaces in Higher Category Theory

In ordinary category theory, we have hom-sets $\mathcal{C}(x, y)$. In enriched category theory, we have hom-objects $\mathcal{C}(x, y)$ in some ambient category $\mathcal{V}$. In higher category theory, we have morphism *spaces* $\mathrm{Map}_{\mathcal{C}}(x, y)$. The definition of these objects is simple in topological or simplicial categories, in fact it is defined as part of the formalism. However, quasi-categories do not contain a definition of mapping spaces as part of their definition, so we need to come up with such a definition.

First, we pause to clear up some misconceptions I have been carrying with me. The homotopy category of a simplicial category $\mathcal{C}$ is computed by considering the image under some (so far unnamed) functor $\mathrm{s}\mathcal{S}\mathrm{et} \to \mathcal{H}$. This functor is any choice of fibrant replacement in Quillen's model structure (a choice of a kan complex), viewed as an object in $\mathrm{Ho}\,\mathrm{s}\mathcal{S}\mathrm{et}$ as defined in Quillen's model structure.

We take the fibrant replacement here since we want to quotient by hommotopy, meaning the domain must be cofibrant and the codomain must be fibrant, by standard model category theory ([?]). In the standard model structure on $\mathcal{T}\mathrm{op}$, CW complexes are cofibrant and all spaces are fibrant, so we need only replace the domain cofibrantly. Similarly, in Joyal's model structure, all simplicial sets are cofibrant and quasi-categories are fibrant. Quillen's model structure also has all objects cofibrant, but Kan complexes are fibrant, which are simply a special case of a quasi-category.

Since the hom-sets in the homotopy category of a topological category are its sets of path-components, it would make sense to define the hom-sets of $\mathrm{h}\mathcal{C}$ in the same way. In particular, we must show $\pi_0 \operatorname{Map}_{\mathcal{C}}(x, y) = R \operatorname{Map}_{\mathcal{C}}(x, y)$, where $R : \mathrm{s}\mathcal{S}\mathrm{et} \to \mathcal{H}$ is a fibrant replacement. Note that the set of path-components of a simplicial set are defined in the expected way: given a composable pair of vertices of the mapping space, a homotopy is a 1-simplex witnessing their composite. We then identify homotopic 1-simplices. Since $R \operatorname{Map}_{\mathcal{C}}(x, y)$ is a Kan complex, this indeed follows: hom-sets in the underlying category are maps $\Delta^0 \to R \operatorname{Map}_{\mathcal{C}}(x, y)$ in $\mathcal{H}$. In $\mathrm{s}\mathcal{S}\mathrm{et}$, these would be choices of vertices, but since $\mathcal{H}$ identifies homotopic maps, this instead becomes a choice of vertices up to homotopy.

Continuing, recall that mapping spaces of the homotopy category of $\mathfrak{C}[S]$ are Kan complexes. We may thus treat $\mathfrak{C}$ as a sort of fibrant replacement for simplicial sets, but only up to homotopy type and on what we expect their mapping spaces to be. An explicit construction of $\mathfrak{C}[S]$ that I will probably do later shows that the mapping spaces are indeed defined in the way we would expect (0-simplices are morphisms between vertices in the sense detailed for quasi-categories, etc). Thus, we would expect the mapping set $\mathrm{h}S(x, y)$ for a simplicial set S to indeed be the hom-set $\mathrm{h}\mathcal{C}[S](x, y)$. Thisi justifies our preivous arguments that I just now realized I didn't understand well. With this fixed understanding, we continue.

Definition 1.2.2.1. Let S be a simplicial set, and let $x, y \in S$ and $\mathcal{H}$ the homotopy category of spaces. Define $\operatorname{Map}_S(x, y) = \operatorname{Map}_{\mathrm{h}S}(x, y) \in \mathcal{H}$.

Of course, when we considered the homotopy category of a simplicial category as $\mathcal{H}$-enriched, we were speaking in the sense of its underlying category being equivalent to the expected definition. However, objects therein are still simplicial sets, they still retain enough structure to be considered a mapping space, and in particular by proposition 1.1.5.7, they are Kan complexes, our models for ∞-groupoids. Since identifying their homotopic vertices gives us our standard notion of the hom-set of the homotopy space, we would expect that the object itself can serve as a mapping space for the simplicial set.

Remark 1.2.2.2. Lurie has the most roundabout motivation possible. I fully understand defining the mapping spaces of simplicial sets or simplicial categories via path-components, but $\mathcal{H}$-objects just make it needlessly confusing in my opinion.

The next question is *how do we compute these mapping spaces*? Not only is computing $\operatorname{Map}_{\mathfrak{C}[S]}(x, y)$ complex, but it need not be a Kan complex in general (only its corresponding homotopy hom-set is), making this entire idea somewhat poorly-founded. To fix this, we simply replace $\operatorname{Map}_{\mathfrak{C}[S]}(x, y)$ with a Kan complex of the same homotopy type. There is a lot of model category theory going on in the background to arrive at these definitions, but luckily we can ignore it!

Define $\operatorname{Hom}^{\mathrm{R}}_S(x, y)$ as the simplicial set whose n-simplices are morphisms $\Delta^{n+1} \to S$ that restrict $z|\Delta^{\{n+1\}} = y$ and $z|\Delta^{0,\dots,n}$ is constant at x, or more expicitly, identifying these restrictions under the Yoneda lemma $\Delta^{\{n+1\}} \cong \Delta^0$, $\Delta^{\{0,\dots,n\}} \cong \Delta^n$ gives the vertex y and the degenerate n-simplex on the vertex x, respectively. A more explicit description is n-simplices of $\operatorname{Hom}^{\mathrm{R}}_S(x, y)$ are $(n+1)$-simplices of S whose $(n+1)$th vertex is y, and all other vertices are x; in particular its $(n+1)$th face is the degenerate n-simplex generated by x. Faces and degeneracies are then defined the same way they are in $S_{\bullet+1}$. We call $\operatorname{Hom}^{\mathrm{R}}_S(x, y)$ the space of *right morphisms* from x to y. Of course, we must still show it is indeed a space.

Proposition 1.2.2.3. *Let $\mathcal{C}$ be an ∞-category. Then for any $x, y \in \mathcal{C}$, the simplicial set* $\operatorname{Hom}^{\mathrm{R}}_{\mathcal{C}}(x, y)$

is a Kan complex.

Proof. By definition 1.2.5.1, which will be proven later, it suffices to show $M = \mathrm{Hom}^{\mathrm{R}}_{\mathcal{C}}(x, y)$ admits fillers for horns $\Lambda^n_i \to M$ with $0 < i \leq n$. Of course, since $n \geq 1$ here, we have by way of proposition 1.1.2.12 that any horn in M is a horn in $\mathcal{C}$, and in particular since the n-simplices of M inject into the $(n+1)$-simplices of $\mathcal{C}$, the nth horn Λ^n_n corresponds to the horn Λ^{n+1}_n in $\mathcal{C}$, rendering all horns inner. It suffices to show the given filler is indeed a simplex of M.

Indeed, let σ fill a horn $\sigma_0 : \Lambda^n_i \to M$, and let τ be the corresponding $(n+1)$-simplex of M. Denote by τ_j the image $\sigma_0(d^j) \in M_n \subseteq \mathcal{C}_{n+1}$ for $j \neq i$. Then $d_j(\tau) = \tau_j$, and since d_j preserves all vertices other than j, and since $n + 1 \geq 2$, tracing the reindexing of vertices that happens when you take a face gives that τ is an n-simplex of M. ■

Unfortunately, there is no obvious composition law on the right morphism space. This is the main drawback of this space, instead of using $\mathrm{Map}_{\mathfrak{C}[S]}(x, y)$. We will show that when S is an ∞-category, they are canonically isomorphic in the homotopy category, concluding that $\mathrm{Hom}^{\mathrm{R}}_S(x, y)$ represents $\mathrm{Map}_S(x, y)$ when S is an ∞-category. This will not be done in this section, since the proof is technical, and will be delegated to chapter 2. For now, we take this statement as fact, along with the fact that a composition law exists up to a contractible space of choices, and make some observations.

Remark 1.2.2.4. $\mathrm{Hom}^{\mathrm{R}}_S(x, y)$ is not self-dual. We define $\mathrm{Hom}^{\mathrm{L}}_S(x, y) \coloneqq \mathrm{Hom}^{\mathrm{R}}_{S^{\mathrm{op}}}(x, y)$, meaning n-simplicies of the *left* mapping space from x to y have their 0th vertex as x, and other vertices as y.

As noted in remark 1.2.2.4, $\mathrm{Hom}^{\mathrm{L}}_S(x, y)$ and $\mathrm{Hom}^{\mathrm{R}}_S(x, y)$ are not in general isomorphic, but they are homotopy equivalent whenever S is an ∞-category. We show this by defining a self-dual space $\mathrm{Hom}_S(x, y) = \{x\} \times_S S^{\Delta^1} \times_S \{y\}$. Recall the internal hom $S^{\Delta^1}_\bullet = \mathrm{sSet}(\Delta^\bullet \times \Delta^1, S)$. An n-simplex is (for some reason) also described by a map $f : \Delta^n \times \Delta^1 \to S$ which restricts along $f|\Delta^n \times \{0\}$ to be constant at x, and $f|\Delta^n \times \Delta^1$ constant at y. With this understanding, we have obvious natural inclusions

$$\mathrm{Hom}^{\mathrm{R}}_S(x, y) \hookrightarrow \mathrm{Hom}_S(x, y) \hookleftarrow \mathrm{Hom}^{\mathrm{L}}_S(x, y).$$

We (much) later show that these inclusions are homotopy equivalences, provided S is an ∞-category.

1.2.3 The Homotopy Category

The nerve N of a category $\mathcal{C}$ is an ∞-category, as noted in notation 1.1.2.4. Informally, we say that N is a fully faithful inclusion from the bicategory $\mathcal{C}\mathrm{at}$ to the $(\infty, 2)$-category $\mathcal{C}\mathrm{at}_\infty$ of ∞-categories. However, we can in fact say more.

Proposition 1.2.3.1. *The nerve functor* $\mathcal{C}\mathrm{at} \to \mathrm{sSet}$ *is right adjoint to the functor* $\mathrm{h} : \mathrm{sSet} \to \mathcal{C}\mathrm{at}$ *which maps simplicial sets* $S \mapsto \mathrm{h}S$*, viewed as the underlying category rather than the* $\mathcal{H}$*-enriched category.*

Proof. We temporarily modify our notation as follows: denote by $N : \mathcal{C}\mathrm{at} \to \mathrm{sSet}$ the nerve, and denote by $N' : \mathrm{s}\mathcal{C}\mathrm{at} \to \mathrm{sSet}$ the simplicial nerve. We can relate these functors by noting that any category $\mathcal{C}$ is viewed as a simplicial category by taking mapping spaces $\mathrm{Map}_{\mathcal{C}}(x, y)$ to be the discrete simplicial set on $\mathcal{C}(x, y)$. This givess an inclusion functor $i : \mathcal{C}\mathrm{at} \hookrightarrow \mathrm{s}\mathcal{C}\mathrm{at}$. The simplicial nerve $N'i(\mathcal{C})$ takes as n-simplices simplicial functors $\mathfrak{C}[\Delta^n] \to i(\mathcal{C})$. We would like to show that the composite $N' \circ i = N$, which we do by showing simplicial functors $\mathfrak{C}[\Delta^n] \to i(\mathcal{C})$ are functors $[n] \to \mathcal{C}$.

Indeed, a simplicial functor $F : \mathfrak{C}[\Delta^n] \to i(\mathcal{C})$ is a choice of n objects, and an n-simplex $\tau \in \mathrm{Map}_{i(\mathcal{C})}(F(0), F(n))$ whose vertices define a chain of morphisms from $F(0)$ to $F(n)$. Thus,

$$\begin{tikzcd} \mathcal{C}\mathrm{at} \arrow[r, hook, "i"'] \arrow[rr, bend left, "N"] & \mathrm{s}\mathcal{C}\mathrm{at} \arrow[r, "N'"'] & \mathrm{s}\mathcal{S}\mathrm{et}. \end{tikzcd}$$

Recall that the path components functor $\pi_0 : \mathrm{s}\mathcal{S}\mathrm{et} \to \mathcal{S}\mathrm{et}$ is lax monoidal, and promotes to the homotopy functor $\pi_0 = \mathrm{h} : \mathrm{s}\mathcal{C}\mathrm{at} \to \mathcal{C}\mathrm{at}$ for simplicial categories. In particular, since $\pi_0 : \mathrm{s}\mathcal{S}\mathrm{et} \to \mathcal{S}\mathrm{et}$ is left adjoint to the inclusion $i : \mathcal{S}\mathrm{et} \hookrightarrow \mathrm{s}\mathcal{S}\mathrm{et}$, this adjunction promotes to $\mathrm{h} \dashv i : \mathrm{s}\mathcal{C}\mathrm{at} \to \mathcal{C}\mathrm{at}$. Since $\mathfrak{C}[-] \dashv N'$ by construction, composition of adjoints yields a left adjoint $\mathrm{h} \circ \mathfrak{C}[-] \dashv N$. By definition, the composite $\mathrm{h} \circ \mathfrak{C}[-]$ is the homotopy functor for simplicial sets $\mathrm{h} : \mathrm{s}\mathcal{S}\mathrm{et} \to \mathcal{C}\mathrm{at}$. ■

Remark 1.2.3.2. If $\mathcal{C}$ is a simplicial category, then we obtain that $\mathrm{h}\mathcal{C} \simeq \mathrm{h}N(\mathcal{C})$ whenever $\mathcal{C}$ is fibrant in the sense that its mapping spaces are Kan complexes.

Given a simplicial set S, Joyal referred to $\mathrm{h}S$ as the *fundamental category* of S, motivated by the fact that $\mathrm{h}S$ is the fundamental groupoid of S when S is a Kan complex. I believe this is still used somewhat in the literature.

We likely expect that the structure of the homotopy category $\mathrm{h}\mathcal{C}$ of an ∞-category $\mathcal{C}$ should be something along the lines of

- Objects are the same as $\mathcal{C}$;
- Morphisms are homotopy classes of 1-morphisms;

and none of the other higher categorical data needs to be expressed explicitly, since it should be encoded in the behavior of the homotopies themselves. Indeed this will be the description we come up with. In order to properly work through this, we see fit to formally introduce homotopy and composition in quasi-categories. All definitions are in-line with those in the nerve, and I have kind of been writing these notes already knowing what they are, and I might have already written it down, but why not nail it down one more time. First, we recall some basics.

Definition 1.2.3.3. Let $\mathcal{C}$ be an ∞-category.

1. Objects of $\mathcal{C}$ are vertices $\mathcal{C}_0$.
2. 1-morphisms (often simply called morphisms) of $\mathcal{C}$ are edges $f \in \mathcal{C}_1$; equivalently $f : \Delta^1 \to \mathcal{C}$.
3. Given a morphism f in $\mathcal{C}$, the domain of f is $d_1(f)$; equivalently $f(0)$ viewed as a morphism $\Delta^1 \to \mathcal{C}$.
4. Similarly, the codomain of f is $d_0(f)$; equivalently $f(1)$.

We use the usual notation $f : x \to y$ to denote that f is a 1-morphism with domain x and codomain y.

We begin by defining a category $\pi(\mathcal{C})$, which we later show to be equivalent to $\mathrm{h}\mathcal{C}$. Of course, $\mathrm{Obj}\,\pi(\mathcal{C}) = \mathrm{Obj}(\mathcal{C})$. To define morphisms, first we must define *homotopy*.

Definition 1.2.3.4. Let $\mathcal{C}$ be an ∞-category, and let $f, g : x \to y$ be morphisms. A *homotopy* σ from f to g, written $\sigma : f \Rightarrow g$ (or simply $\sigma : f \to g$), is a 2-simplex $\sigma : \Delta^2 \to \mathcal{C}$ with faces

$$d_0(\sigma) = 1_y, \qquad d_1(\sigma) = g, \qquad d_2(\sigma) = f.$$

We depict this with a diagram

$$\begin{tikzcd} & y \arrow[dr, "1_y"] & \\ x \arrow[ur, "f"] \arrow[rr, "g"'] & & y \end{tikzcd}$$

If there is a homotopy from f to g, we say f and g are *homotopic*.

Notation 1.2.3.5. Going off of definition 1.2.3.4, a diagram

$$\begin{tikzcd} & y \arrow[dr, "g"] & \\ x \arrow[ur, "f"] \arrow[rr, "h"'] & & z \end{tikzcd}$$

in a ∞-category $\mathcal{C}$ denotes a 2-simplex $\sigma \in \mathcal{C}_2$ with faces

$$d_0(\sigma) = g, \qquad d_1(\sigma) = h, \qquad d_2(\sigma) = f.$$

We sometimes call homotopies *2-morphisms*. Since we expect to be able to take homotopy classes of morphisms, we should probably show that homotopy is an equivalence relation on morphisms.

Proposition 1.2.3.6. *Let $\mathcal{C}$ be an ∞-category, and let $x, y \in \mathcal{C}$ be objects. Then homotopy is an equivalence relation on the set of morphisms $x \to y$.*

Proof. First, let $f : x \to y$ in $\mathcal{C}$. To show reflexivity, we will show $s_1(f)$ is a homotopy $f \Rightarrow f$. Indeed, by the simplicial identities ([Tal25]),

$$\begin{aligned} &d_0 s_1(f) = s_0 d_0(f) = s_0(y) = 1_y, \\ &d_1 s_1(f) = f, \\ &d_2 s_1(f) = f, \end{aligned}$$

rendering $s_1(f)$ a homotopy from f to f, showing reflexivity.

We now prove an intermediary result, from which symmetry and transitivity will fall out. Let $f, g, h : x \to y$ be morphisms, and let $\sigma : f \Rightarrow g$ and $\sigma' : f \Rightarrow h$ be homotopies. Let $\sigma'' = s_1 s_0(y) : \Delta^2 \to \mathcal{C}$ be the constant map at y. We want to show there is a 3-simplex τ filling a horn defined by $\sigma, \sigma', \sigma''$. Define $\tau_0 : \Lambda^3_1 \to \mathcal{C}$ by

$$d^0 \mapsto \sigma'', \qquad d^2 \mapsto \sigma', \qquad d^3 \mapsto \sigma,$$

and by proposition 1.1.2.12 it suffices to show $d_0(\sigma') = d_1(\sigma'')$ and $d_2(\sigma) = d_2(\sigma')$. Indeed, since σ, σ' are homotopies with domain f, we have

$$d_2(\sigma) = f, \qquad d_2(\sigma') = f.$$

Additionally, the 0th face of any homotopy is the identity, so $d_0(\sigma') = 1_y$. Since $\sigma'' = s_1 s_0(y)$, we have by the simplicial identities

$$d_1(\sigma') = d_1 s_1 s_0(y) = s_0(y) =: 1_y.$$

Therefore, there exists a horn $\tau_0 : \Lambda^3_1 \to \mathcal{C}$ as claimed, and since $\mathcal{C}$ is an ∞-category, there exists a 3-simplex $\tau : \Delta^3 \to \mathcal{C}$ rendering the diagram

$$\begin{tikzcd} \Lambda^3_1 \arrow[r, "\tau_0"] \arrow[d, hook] & \mathcal{C} \\ \Delta^3 \arrow[ur, "\tau"'] & \end{tikzcd}$$

commutative. We now show that $d_1(\tau)$ is a homotopy $g \Rightarrow h$. Indeed, by the simplicial identities,

$$d_0 d_1(\tau) = d_0 d_0(\tau) = 1_y, \qquad d_1 d_1(\tau) = d_1 d_2(\tau) = h, \qquad d_2 d_1(\tau) = d_1 d_3(\tau) = g.$$

We now conclude the proof with a few remarks. First, we take the special case $f = h$ of the above situation. We then have that the hypothesis above is that there is a homotopy from f to g, and the conclusion is that there is thus a homotopy from g to f. This shows symmetry. From this, we derive transitivity from the above situation as follows:

1. Let there be a homotopy $f \Rightarrow g$ and $g \Rightarrow h$.
2. By symmetry, there is a homotopy $g \Rightarrow f$.
3. We may invoke the above result on the homotopies $g \Rightarrow f$ and $g \Rightarrow h$, obtaining a homotopy $f \Rightarrow h$.

■

Notation 1.2.3.7. From now on, due to convenient notation, we may choose to denote a horn $\Lambda^n_i \to \mathcal{C}$ by a pair $(f_0, \ldots, \bullet, \ldots, f_n)$. This notation implies that the horn is defined by $d^j \mapsto f_j$ for $j \neq i$, and the bullet appears in the ith spot to symbolize that there is no map for d^i.

Remark 1.2.3.8. We may also show that homotopy is self-dual. Let $f, g : x \to y$ be homotopic in $\mathcal{C}$, and let σ be a homotopy between them. Consider the diagram

$$\begin{tikzcd} \Lambda^3_2 \arrow[r, "{(\sigma, s_1(f), \bullet, s_0(f))}"] \arrow[d, hook] & \mathcal{C} \\ \Delta^3 \arrow[ur, dotted, "\tau"'] & \end{tikzcd}$$

wherein the dotted line exists since $\mathcal{C}$ is an ∞-category, rendering the diagram commutative. The face $d_2(\tau)$ forms the relevant homotopy in $\mathcal{C}$.

We may now make precice our morphisms in $\pi(\mathcal{C})$ as homotopy classes of morphisms in $\mathcal{C}$. More precisely, denote by $\mathcal{C}(x, y)$ the set of morphisms $x \to y$ in $\mathcal{C}$, and denote by $\sim$ the homotopy relation on $\mathcal{C}(x, y)$. Then $\pi(\mathcal{C})(x, y) \coloneqq \mathcal{C}(x, y)/\!\sim$. Of course, we still need a valid composition operation in $\pi(\mathcal{C})$.

Let $f : x \to y$ and $g : y \to z$ in $\mathcal{C}$, and since $d_1(g) = d_0(f)$, by proposition 1.1.2.12 there is a horn $(g, \bullet, f) : \Lambda^2_1 \to \mathcal{C}$. This extends to a 2-simplex $\sigma : \Delta^2 \to \mathcal{C}$. Define $[g] \circ [f] = [d_1(\sigma)]$.

Proposition 1.2.3.9. *Let $\mathcal{C}$ be an ∞-category. Then composition in $\pi(\mathcal{C})$ is well-defined. More precisely, the composite $[g] \circ [f]$ does not depend on the choice of representative, nor the choice of filler σ.*

Proof. We start by verifying the independence of σ. Let $\sigma, \sigma' : \Delta^2 \to \mathcal{C}$ be fillers for the horn $(g, \bullet, f) : \Lambda^2_1 \to \mathcal{C}$, and thus

$$d_0(\sigma) = d_0(\sigma') = g, \qquad d_2(\sigma) = d_2(\sigma') = f.$$

We thus have a horn $(s_1(g), \bullet, \sigma', \sigma)$, which admits a filler τ since $\mathcal{C}$ is an ∞-category. By the same logic as the proof of proposition 1.2.3.6, $d_1(\tau)$ is a homotopy from $d_1(\sigma)$ to $d_1(\sigma')$.

We now show homotopy classes don't depend on representative. Since $\sigma_0 = (g, \bullet, f)$ is a horn, let σ be a 2-simplex with faces $d_0(\sigma) = g$ and $d_2(\sigma) = f$. I just want to prove that this is indeed a horn:

$$d_0\sigma_0(d^2) = d_0(g) = y = d_1(f) = d_1\sigma_0(d^0),$$

and it follows by $2 - 1 = 1$ and proposition 1.1.2.12. Let $\sigma' : f \Rightarrow f'$ be a homotopy, for $f' : x \to y$ some morphism in $\mathcal{C}$. As noted in remark 1.2.3.8, by duality it suffices to show $[g] \circ [f] = [g] \circ [f']$. There is a horn

$$\begin{tikzcd} \Lambda^3_1 \arrow[r, "{(s_0(g),\bullet,\sigma,\sigma')}"] \arrow[d, hook] & \mathcal{C} \\ \Delta^3 \arrow[ur, dotted, "\tau"'] & \end{tikzcd}$$

and since $\mathcal{C}$ is an ∞-category, there exists a dotted line as indicated. Define $\sigma'' = d_1(\tau)$. Note that σ'' fills $(g, \bullet, f')$:

$$d_0(\sigma'') = d_0d_1(\tau) = d_0d_0(\tau) = d_0s_0(g) = g,$$

$$d_2(\sigma'') = d_2d_1(\tau) = d_1d_3(\tau) = d_1(\sigma') = f'.$$

Thus, $[g] \circ [f'] = [d_1(\sigma'')]$. However, we also have

$$d_1(\sigma'') = d_1d_1(\tau) = d_1d_2(\tau) = d_1(\sigma),$$

and since $[d_1(\sigma)] = [g] \circ [f]$, we have $[g] \circ [f'] = [g] \circ [f]$. ■

Proposition 1.2.3.10. *Let $\mathcal{C}$ be an ∞-category. Then $\pi(\mathcal{C})$ is a category.*

Proof. At this point, we need only check identities and associativity. Let $x \in \mathcal{C}$, and $f : x \to y$, and recall that the identity 1_x is given by $[s_0(x)]$. We construct the horn $(f, \bullet, 1_x)$, and want to show $s_0(f)$ fills the horn. Indeed, by the simplicial identities,

$$d_0s_0(f) = f, \qquad d_2s_0(f) = s_0d_1(f) = s_0(x) = 1_x.$$

We thus have that $d_1s_0(f) = f$, and by reflexivity, $[f] \circ [1_x] = [f]$.

Now for associativity. Let

$$w \xrightarrow{f} x \xrightarrow{g} y \xrightarrow{h} z$$

be a composable triple in $\mathcal{C}$. By filling the relevant horns, choose 2-simplices $\sigma, \sigma', \sigma'' : \Delta^2 \to \mathcal{C}$ given by the diagrams, in order,

$$\begin{tikzcd} & x \arrow[dr, "g"] & \\ w \arrow[ur, "f"] \arrow[rr, "gf"'] & & y \end{tikzcd}$$

$$\begin{tikzcd} & y \arrow[dr, "h"] & \\ w \arrow[ur, "gf"] \arrow[rr, "h(gf)"'] & & z \end{tikzcd}$$

$$\begin{tikzcd} & y \arrow[dr, "h"] & \\ x \arrow[ur, "g"] \arrow[rr, "hg"'] & & z \end{tikzcd}$$

We claim that $(\sigma'', \sigma', \bullet, \sigma)$ is a horn $\Lambda^3_2 \to \mathcal{C}$. Indeed,

$$d_0(\sigma') = h = d_0(\sigma''), \qquad d_1(\sigma) = gf = d_2(\sigma').$$

Thus, there is a 3-simplex $\tau : \Delta^3 \to \mathcal{C}$ filling the horn $(\sigma'', \sigma', \bullet, \sigma)$. Note that $[h] \circ [gf] = [d_1(\sigma')]$ by construction. Define $\sigma''' := d_2(\tau)$, and note that

$$d_1(\sigma''') = d_1 d_2(\tau) = d_1 d_1(\tau) = d_1(\sigma').$$

It thus suffices to show σ''' witnesses the composite of hg and f. Indeed,

$$d_0(\sigma''') = d_0 d_2(\tau) = d_1 d_0(\tau) d_1(\sigma'') = hg,$$

$$d_2(\sigma''') = d_2 d_2(\tau) = d_2 d_3(\tau) = d_2(\sigma) = f,$$

and since $d_0(f) = d_1(hg)$, there is a horn $(hg, \bullet, f)$ filled by σ''', concluding the proof. ■

Remark 1.2.3.11. To show that $\pi(\mathcal{C}) \cong \mathrm{h}\mathcal{C}$, we note that proposition 1.2.3.1 allows us to view $\mathrm{h}\mathcal{C}$ in the following way:

1. Objects are vertices of $\mathcal{C}$;
2. For every edge f of $\mathcal{C}$, there is a morphism $f : d_1(f) \to d_0(f)$ in $\mathrm{h}\mathcal{C}$;
3. For each 2-simplex $\sigma : \Delta^2 \to \mathcal{C}$, there is an equality $d_0(\sigma) \circ d_2(\sigma) = d_1(\sigma)$;
4. For each vertex x, the morphism $s_0(x)$ is the identity 1_x.

Intuitively, this description is the same for the following reason: let $f \sim g$ be homotopic. Then there should be a 2-simplex $d_0(\sigma) = 1$, $d_1(\sigma) = g$, and $d_2(\sigma) = f$. However, we note that since $[1] \circ [f] = [f]$, and since this is witnessed by another 2-simplex with $d_0(\sigma') = 1$, $d_1(\sigma') = f$, and $d_2(\sigma') = f$, we have that $f = g$ in $\mathrm{h}\mathcal{C}$ by (3). This leads us to the conclusion of this section.

Proposition 1.2.3.12. *Let $\mathcal{C}$ be an ∞-category. Then there is a functor $F : \mathrm{h}\mathcal{C} \to \pi(\mathcal{C})$ which is the identity on objects, and maps morphisms by $f \mapsto [f]$. Moreover, F is an* isomorphism *of categories.*

Proof. By way of proposition 1.2.3.9 and remark 1.2.3.11, the functor clearly exists, and is manifestly bijective on objects and surjective on morphisms. It suffices to show the functor is faithful.

In remark 1.2.3.11, we stated that every morphism in $\mathcal{C}$ gives rise to a morphism in $h\mathcal{C}$ as described therein. We must start by showing the converse; in particular we must show the composite of two morphisms is a morphism, and that it contains identities. The second statement is trivial, since $s_0(x)$ is an edge of $\mathcal{C}$, so we focus on the first. By logic discussed at length previously which I make precise in proposition 1.2.3.13 because I can, if $f : x \to y$ and $g : y \to z$ in $\mathcal{C}$, then there is a 2-simplex

$$\begin{tikzcd} & y \arrow[dr, "g"] & \\ x \arrow[ur, "f"] \arrow[rr, "gf"'] & & z \end{tikzcd}$$

Thus, any composite in $h\mathcal{C}$ gives rise to this simplex, and since gf is an edge in $\mathcal{C}$, the statement is shown.

Now let $f, g : x \to y$ be homotopic. By remark 1.2.3.11, they are equal in $h\mathcal{C}$, showing F is faithful. ∎

Proposition 1.2.3.13. *Let $\mathcal{C}$ be an ∞-category, and let $f : x \to y$ and $g : y \to z$. Then there is a horn $\sigma_0 \coloneqq (g, \bullet, f) : \Lambda^2_1 \to \mathcal{C}$.*

Proof. Note that
$$d_0\sigma_0(d^2) = d_0(f) = y = d_1(g) = d_1\sigma_0(d^0).$$
The statement now follows by proposition 1.1.2.12. ∎

1.2.4 Objects, Morphisms, and Equivalences

Generalizing the notion introduced in the previous section, in any model of ∞-category, two morphisms are homotopic if they are equal in the homotopy category. In topological categories, this means they are paths in the same path component of the mapping space $\mathrm{Map}_{\mathcal{C}}(x, y)$. We write $f \sim g$ to say that f is homotopic to g.

Definition 1.2.4.1. Let $\mathcal{C}$ be an ∞-category. A morphism $f : x \to y$ in $\mathcal{C}$ is an *equivalence* if it is an isomorphism in the homotopy category. Equivalently, it has a homotopy inverse $g : y \to x$ with $fg \sim 1$ and $gf \sim 1$. We say objects $x, y \in \mathcal{C}$ are *equivalent* if there is an equivalence between them, meaning they are isomorphic in $h\mathcal{C}$.

If $\mathcal{C}$ is a topological category, we can say some specific things about equivalences.

Proposition 1.2.4.2. *Let $f : x \to y$ in a topological category $\mathcal{C}$. The following are equivalent.*

1. *The morphism f is an equivalence;*
2. *The morphism f has a homotopy inverse $g : y \to x$ with $fg \sim 1$ and $gf \sim 1$;*
3. *For each object $z \in \mathcal{C}$, the induced map $\mathrm{Map}_{\mathcal{C}}(z, x) \to \mathrm{Map}_{\mathcal{C}}(z, y)$ is a homotopy equivalence;*
4. *For each object $z \in \mathcal{C}$, the induced map $\mathrm{Map}_{\mathcal{C}}(z, x) \to \mathrm{Map}_{\mathcal{C}}(z, y)$ is a weak homotopy equivalence;*
5. *For each object $z \in \mathcal{C}$, the induced map $\mathrm{Map}_{\mathcal{C}}(y, z) \to \mathrm{Map}_{\mathcal{C}}(x, z)$ is a homotopy equivalence;*

6. *For each object $z \in \mathcal{C}$, the induced map* $\mathrm{Map}_{\mathcal{C}}(y, z) \to \mathrm{Map}_{\mathcal{C}}(x, z)$ *is a weak homotopy equivalence.*

In general, all meaningful properties of objects are invariant under equivalence, and similarly all meaningful properties of morphisms are invariant under homotopy (2-isomorphism). I have a feeling we will rigorize this later using some kind of ∞-Yoneda. For now, we appeal to the following proposition due to Joyal, which we also can't prove yet.

Proposition 1.2.4.3. *Let $\mathcal{C}$ be an ∞-category and $f : \Delta^1 \to \mathcal{C}$ a morphism of $\mathcal{C}$. Then f is an equivalence if and only if, for each $n \geq 2$ and every map $f_0 : \Lambda^2_0 \to \mathcal{C}$ which restricts as $f_0|\Delta^{\{0,1\}} = f$, there exists a filler for f_0.*

We note that the datum of the restriction $f_0|\Delta^{\{0,1\}}$ is determined by the action of f_0 on the coface $d^2 : [1] \to [2]$.

1.2.5 ∞-Groupoids and Classical Homotopy Theory

There is an interesting, albeit not too suprising, correlation between the theory of ∞-categories and ∞-groupoids.

Definition 1.2.5.1. Let $\mathcal{C}$ be an ∞-category. We say $\mathcal{C}$ is an ∞-*groupoid* if the homotopy category $\mathrm{h}\mathcal{C}$ is a groupoid.

We may now make precise the fact that the theory of ∞-groupoids is the same as classical homotopy theory, by way of the following proposition, also due to Joyal.

Proposition 1.2.5.2 ([Joy08a])**.** *Let $\mathcal{C}$ be a simplicial set. The following are equivalent.*

1. *$\mathcal{C}$ is an ∞-groupoid in the sense of definition 1.2.5.1.*
2. *$\mathcal{C}$ admits fillers for all horns Λ^n_i with $0 \leq i < n$.*
3. *$\mathcal{C}$ admits fillers for all horns Λ^n_i with $0 < i \leq n$.*
4. *$\mathcal{C}$ is a Kan complex, meaning it fills all horns.*

Proof. By proposition 1.2.4.3, a 0th horn maps $d^n \mapsto f$ with f an equivalence if and only if it admits a filler. Since all maps in an ∞-groupoid are equivalences, proposition 1.2.4.3 witnesses $(1 \iff 2)$. Duality gives $(1 \iff 3)$. We conclude by noting $(4 \iff 2 \wedge 3)$. ∎

The statement that the theory of ∞-groupoids is the same as classical homotopy theory then reduces to the Quillen adjunction $|-| \dashv \mathrm{Sing}$, and the model structures of $\mathcal{CG}$ and $\mathrm{s}\mathcal{S}\mathrm{et}$. This is much easier parsed using the theory of quasi-categories than some other formulations. For example, proving this with the theory of topological categories requires an adjunction $\Omega \dashv B$ between the loop space and the classifying space of the topological monoid.

Informally, we would say the inclusion functor i from the full subcategory of Kan complexes into the ∞-category $\mathcal{C}\mathrm{at}_\infty$ exhibits ∞-$\mathcal{G}\mathrm{rpd}$ as a full subcategory of the $(\infty, 2)$-category $\mathcal{C}\mathrm{at}_\infty$. There is a converse to this operation, often called the *core* of an ∞-category.

Proposition 1.2.5.3 ([Joy08b])**.** *Let $\mathcal{C}$ be an ∞-category, and let $\mathrm{Core}(\mathcal{C}) \subseteq \mathcal{C}$ be the largest simplicial subset which forms an ∞-groupoid. Then $\mathrm{Core}(\mathcal{C})$ is a Kan complex, and is universal in the sense that for any Kan complex K, postcomposition by the inclusion $i^* : \mathrm{s}\mathcal{S}\mathrm{et}(K, \mathrm{Core}(\mathcal{C})) \to \mathrm{s}\mathcal{S}\mathrm{et}(K, \mathcal{C})$ is an isomorphism.*

Proof. Proposition 1.2.5.2 proves that $\operatorname{Core}(\mathcal{C})$ is a Kan complex. To prove existence, see construction 1.2.5.4. To prove universality, any morphism of simplicial sets $f : K \to \mathcal{C}$ necessarily preserves equivalences since $\mathrm{h}f : \mathrm{h}K \to \mathrm{h}\mathcal{C}$ is a functor and preserves isomorphisms. Thus, it factors uniquely through the inclusion $i : \operatorname{Core}(\mathcal{C}) \hookrightarrow \mathcal{C}$. ■

Construction 1.2.5.4 ([Lur25, 01CZ])**.** Let $\mathcal{C}$ be an ∞-category. Let $\operatorname{Core}(\mathcal{C})$ be the simplicial subset whose n-simplices are preciely those maps $\Delta^n \to \mathcal{C}$ whose edges are all equivalences. This is a simplicial subset by construction, and satisfies the universal property of proposition 1.2.5.3 by construction.

The inclusion $\infty\text{-}\mathcal{G}\mathrm{rpd} \hookrightarrow \mathcal{C}\mathrm{at}_\infty$ doesn't have a left adjoint explicitly, but it has one in a higher categorical sense, which is given by any fibrant replacement with respect to Quillen's model structure. For example, $\operatorname{Sing}|-|$ suffices.

Remark 1.2.5.5. Lurie makes a fascinating remark here. The *identity* functor $1 : \mathrm{s}\mathcal{S}\mathrm{et} \to \mathrm{s}\mathcal{S}\mathrm{et}$ is a Quillen adjunction between the Quillen model structure and the Joyal model structure, which we will define later. In retrospect this is obvious, but I still find it fascinating.

1.2.6 Homotopy Commutativity versus Homotopy Coherence

There is a problem in higher category theory introduced by the notion of a commutative diagram. The main difference between diagrams in $\mathrm{h}\mathcal{C}$ and $\mathcal{C}$ is that one asks whether composites are *equal* in the former, and *homotopic* in the latter. Thus, we must consider the additional datum of a homotopy. The notion of a commutative diagram in $\mathrm{h}\mathcal{C}$ gives rise to a *homotopy commutative* diagram in $\mathcal{C}$, which turns out to be fairly unnatual.

To understand the problem, we pass to the theory of topological categories and let $F : \mathcal{J} \to \mathcal{H}$ be a diagram in the homotopy category of spaces, with $\mathcal{J}$ an ordinary category. We may view $\mathcal{J}$ as a topological category via discretization. Defining a homotopy commutative diagram in $\mathcal{CG}$ corresponding to the diagram in $\mathcal{H}$ requires lifting F to a functor $\widetilde{F} : \mathcal{J} \to \mathcal{CG}$ such that the functor $\mathcal{J} \to \mathcal{H}$ induced by $\widetilde{F}$ is naturally isomorphic to F. This is not always possible.

To see why this is the case, assume such a lift exists, and let $f : x \to y$ and $g : y \to z$ be morphisms in the diagram F. The condition that the functor induced by $\widetilde{F}$ and F be naturally isomorphic implies that there are homotopies $k_f : \widetilde{F}(f) \simeq F(f)$. One then obtains a canonical homotopy

$$F(gf) \simeq \widetilde{F}(gf) = \widetilde{F}(g)\widetilde{F}(f) \simeq F(g)F(f).$$

The functor F is really just a first approximation of $\widetilde{F}$, and the second approximation should be given by accounting for the homotopies as described above. These homotopies cannot be arbitrary; even just accounting for associativity requires a relationship exist between the homotopies. This data can be encoded by the existence of a hiher homotopy, and so on. This is the idea of a *homotopy coherent diagram*: the datum of one should consist of a functor $F : \mathcal{J} \to \mathrm{h}\mathcal{C}$, together with *all* the higher homotopic data given by a lift $\widetilde{F} : \mathcal{J} \to \mathcal{C}$. This is a big issue in higher category theory, since this is a lot of information to specify, and can be hard to work with.

Quasi-categories, once again, provide a neat formulation of this topic. A diagram in a quasi-category $\mathcal{C}$ is a morphism of simplicial sets $f : N(\mathcal{J}) \to \mathcal{C}$, where $\mathcal{J}$ is an ordinary category. We often abuse notation and write $f : \mathcal{J} \to \mathcal{C}$, with the understanding that we identify 1-categories with their nerves in the theory of ∞-categories.

1.2.7 Functors Between Higher Categories

The notion of a homotopy coherent diagram is a special case of a functor of ∞-categories, sometimes simply called an ∞-functor or an $(\infty, 1)$-functor.

Definition 1.2.7.1. Let $\mathcal{C}, \mathcal{D}$ be ∞-categories. An ∞-functor F from $\mathcal{C}$ to $\mathcal{D}$ is a morphism $F : \mathcal{C} \to \mathcal{D}$ of simplicial sets.

Once again, we see that the theory of quasi-categories provides a very succinct model for ∞-functors. Of course, we would like to construct an ∞-category $\mathrm{Fun}(\mathcal{C}, \mathcal{D})$. We propose the following definition:

Definition 1.2.7.2. Let $\mathcal{C}, \mathcal{D}$ be simplicial sets (athough we only use this notation when $\mathcal{D}$ is an ∞-category). We let $\mathrm{Fun}(\mathcal{C}, \mathcal{D})$ denote the simplicial set given by the internal hom $\mathrm{Map}_{\mathrm{s}\mathcal{S}\mathrm{et}}(\mathcal{C}, \mathcal{D})_n = \mathrm{s}\mathcal{S}\mathrm{et}(\Delta^n \times \mathcal{C}, \mathcal{D})$.

Proposition 1.2.7.3. *Let K be a simplicial set.*

1. *For every ∞-category $\mathcal{C}$, the simplicial set $\mathrm{Fun}(K, \mathcal{C})$ is an ∞-category.*
2. *Let $\mathcal{C} \to \mathcal{D}$ be a categorical equivalence (definition 1.1.5.11) of ∞-categories. Then the induced map $\mathrm{Fun}(K, \mathcal{C}) \to \mathrm{Fun}(K, \mathcal{D})$ is a categorical equivalence.*
3. *Let $\mathcal{C}$ be an ∞-category and $K \to K'$ a categorical equivalence of simplicial sets. Then the induced map $\mathrm{Fun}(K', \mathcal{C}) \to \mathrm{Fun}(K, \mathcal{C})$ is a categorical equivalence.*

We delegate the proof till later, since it uses Joyal's model structure. Morphisms in $\mathrm{Fun}(\mathcal{C}, \mathcal{D})$ are called *natural transformations*, and equivalences are *natural equivalences*. We could of course append ∞ to the beginning, but "∞-natural transformation" just looks stupid in my opinion.

1.2.8 Joins of ∞-Categories

Let $\mathcal{C}$ and $\mathcal{D}$ be *ordinary 1-categories*. We can define a new category $\mathcal{C} \star \mathcal{D}$, called the *join* of $\mathcal{C}$ and $\mathcal{D}$, defined by

$$\mathrm{Obj}(\mathcal{C} \star \mathcal{D}) = \mathrm{Obj}(\mathcal{C}) \amalg \mathrm{Obj}(\mathcal{D}),$$

and for each $x, y \in \mathcal{C} \star \mathcal{D}$,

$$(\mathcal{C} \star \mathcal{D})(x, y) = \begin{cases} \mathcal{C}(x, y), & x, y \in \mathcal{C} \\ \mathcal{D}(x, y), & x, y \in \mathcal{D} \\ \varnothing, & x \in \mathcal{D}, y \in \mathcal{C} \\ *, & x \in \mathcal{C}, y \in \mathcal{D} \end{cases}.$$

Apparently this construction is useful when discussing diagrams, limits, and colimits. We will generalize it to the ∞-categorical setting in this section.

Definition 1.2.8.1. Let S, S' be simplicial sets. Define the *join* of S and S', denoted $S \star S'$, as follows: for each nonempty finite linearly ordered set J, set

$$(S \star S')(J) = \coprod_{J = I \cup I'} S(I) \times S'(I'),$$

where the coproduct is computed over all decompositions of J into *disjoint* subsets I, I', satisfying $\max I < \min I'$. We take the convention that $S(\varnothing) = S'(\varnothing) = *$.

Restricting along $\mathbf{\Delta}$, we obtain a more explicit description

$$(S \star S')_n = S_n \amalg S'_n \amalg \coprod_{i+j=n-1} S_i \times S'_j.$$

It's not immediately clear why these are equivalent, but Lurie explains more in [Lur25, 016K]. The empty simplicial set $\varnothing = \Delta^{-1}$ is an identity for the operation:

$$(S \star \varnothing)_n = S_n \amalg \varnothing \amalg \coprod_{i+j=n-1} S_i \times \varnothing = S_n.$$

In fact, $\star$ induces a monoidal structure on $\mathrm{s}\mathcal{S}\mathrm{et}$ in the obvious way. We also have natural isomorphisms $\varphi_{ij} : \Delta^{i-1} \star \Delta^{j-1} \cong \Delta^{i+j-1}$ for all $i, j \geq 0$.

Remark 1.2.8.2. This is an important point. Using our equivalent definition, we can describe the n-simplices of the join as the set of all tuples (σ, τ) with $\sigma \in S_i$ and $\tau \in S'_j$, where $i + j = n - 1$. One might object that we also have S_n and S'_n in the disjoint union, but if we allow $i, j \geq -1$ and recall that $[-1] = \varnothing$ and we have taken the convention that $S(\varnothing) = S'(\varnothing) = *$, we obtain that $S_{-1} = S'_{-1} = *$.

The main idea behind the join is something inbetween a union and a coproduct. We disjointly insert both objects, but then we connect them together.

The nerve respects joins. Let $\mathcal{C}, \mathcal{D}$ be (1-, simplicial, topological) categories. Then there is a canonical isomorphism

$$N(\mathcal{C}) \star N(\mathcal{D}) \cong N(\mathcal{C} \star \mathcal{D})$$

for N the (regular, simplicial, topological) nerve. I assume this is not terribly difficult to prove, but it seems tedious. This of course suggests that the proposed join on simplicial sets (and in particular, ∞-categories) is the same as that on categories. Unfortunately, $\mathfrak{C}[\bullet]$ does not commute with $\star$ up to isomorphism, but we will later show that it commutes up to equivalence of simplicial categories, with the equivalence functor the identity on both arguments.

We conclude this section with another result due to Joyal, and some terminology that we will use later.

Proposition 1.2.8.3. *Let S, S' be ∞-categories. Then their join $S \star S'$ is an ∞-category.*

Proof. Let $p : \Lambda^n_i \to S \star S'$ be a horn. If all vertices are carried into either S or S', p has a filler by hypothesis, so we may freely assume there is some $j \in [n]$, $j \neq i$ such that $p : \{0, \dots, j\} \to S_0$ and $p : \{j+1, \dots, n\} \to S'_0$. Of course p then restricts along these vertices by

$$\Delta^{\{0,\dots,j\}} \to S, \qquad \Delta^{\{j+1,\dots,n\}} \to S,$$

and this determines an extension of p mapping $\Delta^n \to S \star S'$. ■

Definition 1.2.8.4. Let K be a simplicial set. The *left cone* $K^{\triangleleft}$ is the join $\Delta^0 \star K$, and the *right cone* $K^{\triangleright}$ is the join $K \star \Delta^0$.

1.2.9 Overcategories and Undercategories

The most important examples of comma categories are the slice categories $(\mathcal{C} \downarrow x) = \mathcal{C}/x$ and coslice categories $(x \downarrow \mathcal{C}) = x/\mathcal{C}$. The text denotes these by $\mathcal{C}_{/x}$ and $\mathcal{C}_{x/}$, and we inherit the books notation for now. These categories are also called *overcategories* and *undercategories* for obvious reasons.

The overcategory can be restated with a universal property: for any category $\mathcal{C}'$, there is an isomorphism of *sets*

$$\mathrm{Hom}(\mathcal{C}', \mathcal{C}_{/x}) \cong \mathrm{Hom}_x(\mathcal{C}' \star [0], \mathcal{C}),$$

where the subscript x denotes that we consider only those functors $\mathcal{C}' \star [0] \to \mathcal{C}$ whose restriction to $[0]$ is x. This property will be used to generalize overcategories to the ∞-categorical setting. This generalization is due to Joyal.

Proposition 1.2.9.1. *Let S, K be simplicial sets, and $p : K \to S$ a morphism of simplicial sets. Then there is a simplicial set $S_{/p}$ satisfying the universal property*

$$\mathrm{s}\mathcal{S}\mathrm{et}(Y, S_{/p}) \cong \mathrm{s}\mathcal{S}\mathrm{et}_p(Y \star K, S).$$

Once again, $\mathrm{s}\mathcal{S}\mathrm{et}_p$ denotes that we only consider maps $f : Y \star K \to S$ with $f|K = p$.

Proof. We prove by explicit construction of $S_{/p}$. Simply define the n-simplices to be the set $\mathrm{s}\mathcal{S}\mathrm{et}_p(\Delta^n \star K, S)$. Of course if $Y = \Delta^n$ the conclusion follows by Yoneda. I think the general proof follows by noting that $- \star K$ preserves colimits, and then doing functors as colimits of representable functors. ■

If S is an ∞-category, we call $S_{/p}$ an *overcategory* of S over p. Passing operations to over/undercategories is well-behaved:

Proposition 1.2.9.2. *Let $p : K \to \mathcal{C}$ be a morphism of simplicial sets with $\mathcal{C}$ and ∞-category. Then $\mathcal{C}_{/p}$ is an ∞-category. Moreover, given a categorical equivalence $q : \mathcal{C} \simeq \mathcal{C}'$, the induced map $\mathcal{C}_{/p} \to \mathcal{C}_{/qp}$ is a categorical equivalence.*

Once again, we cannot do this proof yet. The first chapter is more of an overview of the language; the main machinery is postponed for the rest of the text.

Notation 1.2.9.3. If $\mathcal{C}$ is an ∞-category and $p : \Delta^n \to \mathcal{C}$ is an n-simplex $\sigma \in \mathcal{C}_n$, we generally prefer to write $\mathcal{C}_{/\sigma}$ for the overcategory, for obvious reasons. In particular, with $x \in \mathcal{C}_0$ an object, we write $\mathcal{C}_{/x}$ for the overcategory for the map $\Delta^0 \to \mathcal{C}$ corresponding to the 0-simplex x under Yoneda.

Remark 1.2.9.4. We define the undercategory by noting that $Y \star K$ is dual to $K \star Y$, and thus we obtain for some $p : K \to S$ a morphism of simplicial sets, the *undercategory* $S_{p/}$ is given by the simplicial set which is universal in the sense that

$$\mathrm{s}\mathcal{S}\mathrm{et}(Y, S_{p/}) \cong \mathrm{s}\mathcal{S}\mathrm{et}_p(K \star Y, S).$$

Of course, the definition is simply dual to the one given in proposition 1.2.9.1.

Remark 1.2.9.5 ([nLa25])**.** Another definition for the over or undercategory is as follows. Let $f : K \to \mathcal{C}$ be a morphism of simplicial sets. Define $\mathcal{C}_{/f}$ as the simpliical set which is universal in the sense that

$$\mathrm{s}\mathcal{S}\mathrm{et}(S, \mathcal{C}_{/f}) \cong (K \downarrow \mathrm{s}\mathcal{S}\mathrm{et})(i_{S,K}, f)$$

for $i_{S,K} : K \to S \star K$ the canonical inclusion. Of course, morphisms in the coslice category are diagrams

$$\begin{tikzcd} K \arrow[r, equal] \arrow[d, "i_{S,K}"'] & K \arrow[d, "f"] \\ S \star K \arrow[r, "\varphi"'] & \mathcal{C} \end{tikzcd}$$

whose commutivity implies that $\varphi \circ i_{S,K} = f$. Note that $\varphi \circ i_{S,K} = \varphi|K$, giving the equivalence to our definition. The undercategory is defined by swapping for $i_{K,S} : K \to K \star S$.

Claim 1.2.9.6. *Let $\mathcal{C}$ be a 1-category. Then there is a (canonical) isomorphism $N(\mathcal{C})_{/x} \cong N(\mathcal{C}_{/x})$ for $x \in \mathcal{C}$.*

Proof. The proof amounts to an inspection of the definition of the undercategory of the nerve; the isomorphism will be obvious after this. Recall that n-simplices of the nerve of the overcategory $N(\mathcal{C}_{/x})$ are simply n-simplices of the nerve of $\mathcal{C}$, followed by a map to x. We will essentially prove that $N(\mathcal{C})_{/x}$ is basically the same.

Recall that $\Delta^{i-1} \star \Delta^{j-1} \cong \Delta^{i+j-1}$. Applying this here gives $\Delta^n \star \Delta^0 \cong \Delta^{n+1}$. Thus, n-simplices of $N(\mathcal{C})_{/x}$ are $(n+1)$-simplices $\Delta^{n+1} \to N(\mathcal{C})$. It suffices at this point to show that the codomain of the final map is x, which would conclude the proof by obviousness. Recall that our simplex $\Delta^{n+1} \to \mathcal{C}$ must restrict along the original Δ^0 to x. This corresponds to the final vertex under the isomorphism (the isomorphism is basically just concatenating vertices), so we have the result as claimed. ■

1.2.10 Fully Faithful and Essentially Surjective Functors

The definition of these is easily given in the context of topological categories: a functor must be fully faithful and essentially surjective on the homotopy category. Equivalently, the functor must have weak homotopy equivalences from any object to its image, and must define weak homotopy equivalences on mapping spaces. This is a fairly obvious definition and at this point really does not require motivation.

Translating to quasi-categories requires a slightly more convoluded description of the definition. Since Lurie does not offer this in HTT, and since I really don't feel like going through the annoying work of translating his definition along $\mathfrak{C}$, $|-|$, Sing, and N, we mainly use [Lur25, 01JG] for this section.

Recall in definition 1.2.2.1 we define the mapping space of an ∞-category by $\mathrm{Map}_{\mathcal{C}}(x, y) \coloneqq \mathrm{Map}_{\mathrm{h}\mathcal{C}}(x, y) = \mathrm{Map}_{\mathrm{h}\mathfrak{C}[\mathcal{C}]}(x, y) \in \mathcal{H}$, viewed as the Kan complex in $\mathcal{H}$, rather than the underlying category. We now propose the following definitions.

Definition 1.2.10.1. Let $\mathcal{C}, \mathcal{D}$ be ∞-categories, and $F : \mathcal{C} \to \mathcal{D}$ an ∞-functor. We say F is *fully faithful* if the induced maps on mapping spaces $\mathrm{Map}_{\mathcal{C}}(x, y) \to \mathrm{Map}_{\mathcal{D}}(F(x), F(y))$ are weak homotopy equivalences of Kan complexes (Quillen's model structure) for each $x, y \in \mathcal{C}$.

Due to the calculation being really painful, I take Lurie at his word that given a functor of 1-categories $F : \mathcal{C} \to \mathcal{D}$, F is fully faithful if and only if $N(F) : N(\mathcal{C}) \to N(\mathcal{D})$ is fully faithful in the sense of definition 1.2.10.1, successfully generalizing the definition.

Definition 1.2.10.2. Let $F : \mathcal{C} \to \mathcal{D}$ be an ∞-functor. Then F is *essentially surjective* if for all $d \in \mathcal{D}$, there is $c \in \mathcal{C}$ and an equivalence $F(c) \simeq d$.

This generalizes the standard notion very obviously. If $F : \mathcal{C} \to \mathcal{D}$ is an essentially surjective 1-functor, then for all $d \in N(\mathcal{D})_0$, there is $c \in N(\mathcal{C})_0$ and an isomorphism $F(c) \cong d$. Since isomorphisms are equivalences in nerves, we observe that $N(F)$ is essentially surjective, and trivially the converse holds.

I postpone developing much of the theory of these objects for later in the book, as I have with most sections in this first chapter. The goal is to understand the *language*, not to be extremely rigorous, nor to fully understand all definitions.

1.2.11 Subcategories of ∞-Categories

Let $\mathcal{C}$ be an ∞-category, and let $\mathrm{h}\mathcal{C}' \subseteq \mathrm{h}\mathcal{C}$ be any subcategory of its homotopy category. We construct an ∞-category $\mathcal{C}'$ by constructing the pullback square in $\mathrm{s}\mathcal{S}\mathrm{et}$:

$$\begin{tikzcd}
\mathcal{C}' \arrow[r] \arrow[d] \arrow[dr, phantom, very near start, "\lrcorner"] & \mathcal{C} \arrow[d, "N\mathrm{h}"] \\
N(\mathrm{h}\mathcal{C}') \arrow[r, "\subseteq"'] & N(\mathrm{h}\mathcal{C})
\end{tikzcd}$$

We say $\mathcal{C}'$ is a *subcategory* of $\mathcal{C}$. Of course it is a subcategory as an ∞-category, but we don't write "∞" to avoid annoying language. To show that $\mathcal{C}'$ is an ∞-category, let $0 < i < n$, and consider a horn $\sigma_0 : \Lambda^n_i \to \mathcal{C}'$. Composing with the arrows $\mathcal{C}' \to \mathcal{C}$ and $\mathcal{C}' \to N(\mathrm{h}\mathcal{C}')$ gives the relevant part of the diagram

$$\begin{tikzcd}
\Delta^n \arrow[rrr, bend left] \arrow[ddr, bend right] & & & \\
& \Lambda^n_i \arrow[ul, hook'] \arrow[r] \arrow[urr, phantom] \arrow[rr, bend left] \arrow[d, bend right] & & \\
& N(\mathrm{h}\mathcal{C}') & &
\end{tikzcd}$$

$$\begin{tikzcd}
\Delta^n \arrow[rrrdd, bend left] \arrow[rrddd, bend right] & & & \\
& \Lambda^n_i \arrow[ul, hook'] \arrow[dr] \arrow[rrd, bend left] \arrow[rdd, bend right] & & \\
& & \mathcal{C}' \arrow[r] \arrow[d] \arrow[dr, phantom, very near start, "\lrcorner"] & \mathcal{C} \arrow[d] \\
& & N(\mathrm{h}\mathcal{C}') \arrow[r] & N(\mathrm{h}\mathcal{C})
\end{tikzcd}$$

Of course since $\mathcal{C}, N(\mathrm{h}\mathcal{C}')$ are ∞-categories, there are fillers as indicated, and since each triangle commutes, the entire diagram commutes. Universality induces a morphism $\Delta^n \to \mathcal{C}'$ (not indicated), filling the horn by commutativity.

We say $\mathcal{C}'$ is full when $\mathrm{h}\mathcal{C}'$ is full as a subcategory of $\mathrm{h}\mathcal{C}$, and we note that $\mathcal{C}'$ is determined by the set of objects $\mathcal{C}'_0 \subseteq \mathcal{C}_0$. We then say $\mathcal{C}'$ is the *full subcategory spanned by* $\mathcal{C}'_0$. The inclusion $\mathcal{C}' \hookrightarrow \mathcal{C}$ is fully faithful in this case. In particular, any fully faithful functor $f : \mathcal{C}'' \to \mathcal{C}$ factors as $\mathcal{C}'' \to \mathcal{C}' \to \mathcal{C}$, where the first map is an equivalence and the second is the inclusion of the full subcategory spanned by $f(\mathcal{C}''_0)$.

1.2.12 Initial and Final Objects

Terminology 1.2.12.1. Lurie calls objects "final", whereas I use the term "terminal."

A natural generalization of final objects arises in the context of simplicial and topological categories.

Definition 1.2.12.2. Let $\mathcal{C}$ be a simplicial or topological category. Then $x \in \mathcal{C}$ is final if $\mathrm{Map}_{\mathcal{C}}(x, -)$ is weakly contractible.

This translates to x being final in $\mathrm{h}\mathcal{C}$, which dualizes in the obvious way. Since this makes reference to $\mathrm{h}\mathcal{C}$, this translates to ∞-categories, but we will need an explicit description.

Definition 1.2.12.3. Let $\mathcal{C}$ be a simplicial set. A vertex $x \in \mathcal{C}_0$ is *strongly final* if the projection $\mathcal{C}_{/x} \to \mathcal{C}$ is a trivial fibration of simplicial sets (since cofibrations are mono maps, trivial fibrations lift against monics).

We investigate the projection $\mathcal{C}_{/x} \to \mathcal{C}$ to give a better description of strongly final objects. Recall that $(\mathcal{C}_{/x})_n = \mathrm{s}\mathcal{S}\mathrm{et}_x(\Delta^n \star \Delta^0, \mathcal{C})$. Since $\Delta^n \subseteq \Delta^n \star \Delta^0$, we can restrict n-simplices along the inclusion to obtain morphisms $\Delta^n \to \mathcal{C}$, and by Yoneda, n-simplices of $\mathcal{C}$. This is how we define the projection: precomposition by the inclusion $i^* : \mathrm{s}\mathcal{S}\mathrm{et}_x(\Delta^n \star \Delta^0 \to \mathcal{C}) \to \mathrm{s}\mathcal{S}\mathrm{et}_x(\Delta^n, \mathcal{C}) \cong \mathcal{C}_n$, where the second $\mathrm{s}\mathcal{S}\mathrm{et}_x$ denotes the set of morphisms whose final vertex is x.

Let $x \in \mathcal{C}$ be strongly final. Then the projection is a trivial Kan fibration. Focusing on the Kan fibration condition, we see that any diagram of solid arrows

$$\begin{tikzcd}
X \arrow[r] \arrow[d] & \mathcal{C}_{/x} \arrow[d] \\
Y \arrow[r] \arrow[ur, dotted] & \mathcal{C}
\end{tikzcd}$$

with the left vertical arrow mono, there is a lift as indicated by the dotted arrow, rendering the diagram commutative. We can make the very interesting remark that this is equivalent to the condition that all maps $f_0 : \partial\Delta^n \to \mathcal{C}$ wherein $(0 \mapsto n) \mapsto x$ extend to maps $\Delta^n \to \mathcal{C}$.

To prove this, consider the case where the mono map is $\partial\Delta^n \hookrightarrow \Delta^n$, which is mono since it is levelwise injective. Suppose lifts

$$\begin{tikzcd}
\partial\Delta^n \arrow[r] \arrow[d, hook] & \mathcal{C}_{/x} \arrow[d] \\
\Delta^n \arrow[r] \arrow[ur, dotted] & \mathcal{C}
\end{tikzcd}$$

exist. Suppose for now that any map $f_0 : \partial\Delta^n \to \mathcal{C}$ which has n-th vertex x factors through $\mathcal{C}_{/x}$. Then the lift diagram ensures commutativity of the diagram

$$\begin{tikzcd}
\partial\Delta^n \arrow[r] \arrow[d, hook] \arrow[rr, bend left, "f_0"] & \mathcal{C}_{/x} \arrow[r, "i_\bullet^*"'] & \mathcal{C} \\
\Delta^n \arrow[ur, dotted, "f"'] \arrow[urr, bend right, "i_\bullet^* \circ f"'] & &
\end{tikzcd}$$

for $i_\bullet^*$ the inclusion. It thus suffices to show such factorizations exist. Indeed, let $f_0 : \partial\Delta^n \to \mathcal{C}$ map $(0 \mapsto n) \mapsto x$, and recall that simplices (σ, τ) in $\mathcal{C}_{/x}$ needed to have x as the final vertex of σ, concluding the proof by factoring along the injection $\Delta^n \hookrightarrow \Delta^n \star \Delta^0$.

Now we show the converse. Of course $\mathcal{C}_{/x} \hookrightarrow \mathcal{C}$ is a fibration since we necessarily have $\Lambda_n^i \hookrightarrow \Delta^n$ factors through $\partial\Delta^n \hookrightarrow \Delta^n$, and horns $\Lambda_i^n \to \mathcal{C}$ have nth vertex x iff it factors through $\mathcal{C}_{/x}$. Since boundaries admit fillers here by hypothesis, all lifts necessarily exist. It suffices to show $\mathcal{C}_{/x}$ is a weak equivalence, which requires simplicial homotopy that I don't know yet LOL.

Proposition 1.2.12.4. *Let $\mathcal{C}$ be an ∞-category with an object y. Then y is strongly final if and only if $\mathrm{Hom}^{\mathrm{R}}_{\mathcal{C}}(-, y)$ is weakly contractible.*

Corollary 1.2.12.5. *Let $\mathcal{C}$ be an ∞-category. Then $y \in \mathcal{C}_0$ is final if and only if it is strongly final.*

Proof. I don't fully understand one direction. For the convese, let y be final. Thus, $\mathrm{Map}_{\mathcal{C}}(x, y)$ is contractible, and since that is the homotopy type of $\mathrm{Hom}^{\mathrm{R}}_{\mathcal{C}}(x, y)$, the latter space is also contractible. By proposition 1.2.12.4, y is strongly terminal. ■

Final objects dualize in the obvious way: x is (strongly) initial if the projection $\mathcal{C}_{x/} \to \mathcal{C}$ is a trivial fibration; in particular any sphere $\partial\Delta^n \to \mathcal{C}$ with starting vertex x admits a filler. The remainder of the arguments dualize in an even more obvious way.

Proposition 1.2.12.6. *Let $\mathcal{C}$ be an ∞-category and $\mathcal{C}'$ the full subcategory spanned by terminal vertices. Then $\mathcal{C}'$ is either empty, or a contractible Kan complex.*

Proof. If $\mathcal{C}'$ is empty then the statement follows. Otherwise, to show $\mathcal{C}'$ is a Kan complex, we show $\mathcal{C}' \to *$ is fibrant, meaning $\mathcal{C}' \to *$ right lifts against spheres $\partial\Delta^n \hookrightarrow \Delta^n$. Equivalently, given any diagram of solid arrows

$$\begin{tikzcd} \partial\Delta^n \arrow[r, "f"] \arrow[d, hook] & \mathcal{C}' \\ \Delta^n \arrow[ur, dotted] & \end{tikzcd}$$

there is a dotted arrow as indicated, rendering the diagram commutative. Indeed, since the nth vertex of f is necessarily terminal, since $\mathcal{C}'_0$ is only terminal objects, the map factors through $\mathcal{C}'_{/f(n)}$. Since $\mathcal{C}'_{/f(n)} \to \mathcal{C}'$ is a trivial fibration by hypothesis, there is a filler $\Delta^n \to \mathcal{C}'_{/f(n)}$, which extends to the relevant filler $\Delta^n \to \mathcal{C}'$.

To show contractibility, we explicitly construct a homotopy from 1 and the constant map $\mathcal{C}' \to \Delta^0 \to \mathcal{C}'$ on an arbitrary terminal object x. First we show $X \times \Delta^1$ is a cylinder object for a simplicial set X. Clearly $X \amalg X \to X \times \Delta^1$ given by mapping the first X to $X \times \{0\}$, and the second to $X \times \{1\}$, is monic. It suffices to show $X \times \Delta^1 \to X$ is a weak equivalence, and thus $|X| \times I \to |X|$ is a weak homotopy equivalence of spaces. Since π_n preserves products for $n \geq 0$, and since I is contractible (i.e. $\pi_n(I) = 0$), we have

$$\pi_n(|X| \times I) \cong \pi_n(X) \times \pi_n(I) \cong \pi_n(X).$$

We thus very simply have that the projection $\pi : X \times \Delta^1 \to X$ is a weak equivalence.

Let $i_0 + i_1$ be the map $\mathcal{C}' \amalg \mathcal{C} \to \mathcal{C} \times \Delta^1$ given above. We must show given any diagram of solid arrows

$$\begin{tikzcd} \mathcal{C}' \arrow[d] \arrow[dr, "i_0"] \arrow[drr, bend left, "f"] & & \\ \mathcal{C}' \amalg \mathcal{C}' \arrow[r, "i_0+i_1"] & \mathcal{C}' \times \Delta^1 \arrow[r, dotted, "H"] & \mathcal{C}' \\ \mathcal{C}' \arrow[u] \arrow[ur, "i_1"] \arrow[urr, bend right, "x \in \mathcal{C}'_0"'] & & \end{tikzcd}$$

there exists a dotted homotopy as indicated, rendering the diagram commutative. come back to alter ■

1.2.13 Limits and Colimits

Definition 1.2.13.1. Let $\mathcal{C}$ be an ∞-category and $p : K \to \mathcal{C}$ a morphism of simplicial sets. A *limit* for p, denoted $\lim p$, is a final object of $\mathcal{C}_{/p}$. A *colimit* for p, denoted $\operatorname{colim} p$, is an initial object of $\mathcal{C}_{p/}$.

We define the limit of p as an object in the overcategory for p, so we need a way to identify it with an object in $\mathcal{C}$. Recall that $\mathcal{C}_{/p}$ has n-simplices given by $\mathrm{s}\mathcal{S}\mathrm{et}_p(\Delta^n \star K, \mathcal{C})$. The 0-simplex we are interested in is a map $\overline{p} : (\Delta^0 \star K =: K^{\triangleleft}) \to \mathcal{C}$ which restricts as $\overline{p}|K = p$. We then abuse notation and write $\overline{p}(\infty) \in \mathcal{C}$ as the limit, where ∞ is the point of the cone at the vertex. Similarly for colimits, we abuse notation via $\overline{p} : K^{\triangleright} \to \mathcal{C}$.

Of course, (co)limits need not be unique, but proposition 1.2.12.6 says that they form a contractible Kan complex (meaning its homotopy category is equivalent to a group) when they exist. It of course follows that they are all *equivalent* to each other, aka isomorphic in the ∞-categorical sense.

Later, we will show this definition of (co)limits agrees with the classical theory of homotopy (co)limits in $\mathcal{T}\mathrm{op}$.

Remark 1.2.13.2. Let $\mathcal{C}'$ be a full sub-∞-category of $\mathcal{C}$, and $p : K \to \mathcal{C}'$ a diagram. Then the overcategory $\mathcal{C}'_{p/}$ is a pullbback $\mathcal{C}' \times_{\mathcal{C}} \mathcal{C}_{p/}$. To prove this, consider the square

$$\begin{tikzcd} \mathcal{C}'_{p/} \arrow[r] \arrow[d] & \mathcal{C}' \arrow[d, hook] \\ \mathcal{C}_{p/} \arrow[r, "\pi"'] & \mathcal{C} \end{tikzcd}$$

Does this square commute? Of course, the category $\mathcal{C}_{p/} \cong \mathrm{s}\mathcal{S}\mathrm{et}_p(K \star \Delta^\bullet, \mathcal{C})$ projects onto $\mathcal{C}$ by precomposing along the inclusion $\Delta^\bullet \hookrightarrow K \star \Delta^\bullet$ and identifying under Yoneda, as indicated by π. The inclusion $\mathcal{C}' \hookrightarrow \mathcal{C}$ is obvious. Consider the map $\mathcal{C}'_{p/} \to \mathcal{C}$ to also be the projection. Note that a morphism $K \star \Delta^\bullet \to \mathcal{C}'$ can postcompose along $\mathcal{C}' \hookrightarrow \mathcal{C}$ to yield an $\bullet$-simplex of $\mathcal{C}_{p/}$, so we have the vertical morphism on the left.

To see that the diagram indeed commutes, we chase simplices somewhat. Let $f : K \star \Delta^n \to \mathcal{C}'$ be an n-simplex of $\mathcal{C}'_{p/}$. Composing along the right yields the n-simplex $f|\Delta^n$, viewed as an element of $\mathcal{C}$. Composing along the left yields the n-simplex $(i \circ f)|\Delta^n$, which is precisely the same since really this is just $i \circ \Delta^n \circ i$, and so is the other one. Thus, the square commutes.

Of course, universality remains to be shown. Let $f : X \to \mathcal{C}'$ and $g : X \to \mathcal{C}_{p/}$ be morphisms of simplicial sets such that the square

$$\begin{tikzcd} X \arrow[r] \arrow[d] & \mathcal{C}' \arrow[d] \\ \mathcal{C}_{p/} \arrow[r] & \mathcal{C} \end{tikzcd}$$

commutes. Consider the map $k : X \to \mathcal{C}'_{p/}$ given by $p \star f$???? defined in the obvious way psure this makes the right triangle commute cuz $\mathcal{C}'_{p/} \to \mathcal{C}'$ identifies under yoneda already, but like what about left ok well it suffices to show that g is basically just f on $\mathcal{C}'$. I'm pretty sure $g = g' \star p$ by universality of coproduct, so just gotta show $g' = f$ after playing with inclusions. Ok yeah basically $f(x) = g(x) \circ \iota$ for $\iota : \Delta^\bullet \hookrightarrow \Delta^\bullet \star K$.

1.3 Ridification of Quasi-Categories

I found Lurie's definition of ridigification $\mathfrak{C}$ (which is actually attributed to Joyal) particularly difficult to parse. However, the 2011 paper [DS09] introduced a new characterization of $\mathfrak{C}$, which is not only simpler to understand than Joyal's rigidification, but does not require computing colimits in $\mathrm{s}\mathcal{C}\mathrm{at}$. In this section, we digress from the main text to explore Dugger and Spivak's rigidification functor, and the theory of necklaces.

1.3.1 Necklaces

Definition 1.3.1.1. A *necklace* is a wedge sum of representable simplicial sets of the form

$$\Delta^{n_0} \vee \Delta^{n_1} \vee \cdots \vee \Delta^{n_k},$$

for $n_i \geq 0$, where we glue the final vertex of Δ^{n_i} to the first vertex of $\Delta^{n_{i+1}}$. A necklace is in *preferred form* if either $n_i \geq 1$ for all i, or if $k = 0$. Each Δ^{n_i} is a *bead* of the necklace. A *joint* of the necklace is an initial or final vertex in one of the component Δ^{n_i}s.

Given a necklace T, write $V_T = T_0$ for the set of vertices, and $J_T \subseteq V_T$ for the set of joints of T. Both of these sets admit total orders by writing $a \leq b$ if there is a directed path $a \to b$ in T. We write α_T for the initial vertex of T and ω_T for the final vertex of T. We define the wedge sum $S \vee T$ of two necklaces S, T in the obvious way, by gluing $\omega_S \sim \alpha_T$.

Each necklace T has a canonical morphism $\partial\Delta^1 \to T$ given by $0 \mapsto \alpha_T$ and $1 \mapsto \omega_T$. We define $\mathcal{N}\mathrm{ec}$ as the full subcategory of $\mathrm{s}\mathcal{S}\mathrm{et}_{*,*} = (\partial\Delta^1 \downarrow \mathrm{s}\mathcal{S}\mathrm{et})$ spanned by those canonical morphisms to necklaces $(\partial\Delta^1 \to T, T)$. It follows that morphisms $f : S \to T$ of necklaces are maps of simplicial sets such that $f(\alpha_S) = \alpha_T$ and $f(\omega_S) = \omega_T$. We may then freely identify $\mathcal{N}\mathrm{ec}$ with a subcategory of $\mathrm{s}\mathcal{S}\mathrm{et}$ via the projection of the comma category.

A *simplex* is a necklace with one bead. A *spine* is a necklace where all beads are Δ^1. Every necklace T has a canonically associated simplex and spine: the representable functor $\Delta[T] = \Delta^{V_T}$ is the *simplex* of T, and the spine $\mathrm{Spi}[T]$ is the longest spine in T. There are inclusions $\mathrm{Spi}[T] \hookrightarrow T \hookrightarrow \Delta[T]$, and the latter inclusion is functorial.

If X is a simplicial set, then there is a relation $\leq$ on its vertices given by $x \leq y$ iff there is a spine T and a map $T \to X$ which restricts along α_T and ω_T to the vertices x and y. This is a complex way of saying there is a path $x \to y$ in X. This relation is transitive and reflexive, but need not be antisymmetric (indeed, it is fully symmetric in Kan complexes).

Definition 1.3.1.2. A simplicial set X is *ordered* if

1. the relation $\leq$ on X_0 is antisymmetric, and
2. a simplex $x \in X_n$ is determined by its vertex sequence $x(0) \leq \ldots \leq x(n)$; i.e. no distinct simplices have the same vertex sequences.

I didn't fully understand (2), so I offer some further discussion. I used an AI chatbot to assist me with this information. The vertex sequence of an n-simplex is clearly just the ordered list of its vertices $(v_0, \ldots, v_n)$. The n-simplex witnesses an ordering $v_i \leq v_{i+1}$, $0 \leq i < n$ since its edges map $v_i \to v_{i+1}$. The condition simply says that no two n-simplices have the same vertices in the same order. Also recall that the ordering on the n-simplex $\sigma : \Delta^n \to X$ is given by the images of the vertices $\{0, \ldots, n\} = \Delta^n_0$.

Definition 1.3.1.3. A map $f : A \to X$ of simplicial sets is a *simple inclusion* if it has the right lifting property with respect to the canonical inclusions $\partial\Delta^1 \hookrightarrow T$ for all necklaces T.

Remark 1.3.1.4. The notion of a simple inclusion simply says that any path starting and ending in A must lie entirely in A. More generally, any necklace $T \to X$ starting and ending in A

$(\alpha_T, \omega_T \in \operatorname{Im} A \hookrightarrow X)$ must be fully contained in A. We can immediately see that this is an inclusion of at least vertices, by noting that given any lifting problem against $\partial\Delta^1 \to \Delta^0$ as indicated by the diagram of solid arrows

$$\begin{tikzcd}
\partial\Delta^1 \arrow[r] \arrow[d] & A \arrow[d, hook] \\
\Delta^0 \arrow[r] \arrow[ur, dotted] & X
\end{tikzcd}$$

admits a dotted arrow as indicated, rendering the diagram commutative. An arrow $\partial\Delta^1 \to A$ is simply a choice of two vertices, and the commutativity of the outer square says that the simple inclusion identifies these two vertices. The existence of the lift says that these two vertices are in fact the *same*, showing that the simple inclusion is injective on vertices.

We would also like to show higher levels of simple inclusions are injective, since we would then have that they are mono. This is a classical theorem of simplicial sets that I have never proven before.

Proposition 1.3.1.5. *A morphism of simplicial sets $f : A \to X$ is monic (levelwise injective) if and only if it has the right lifting property with respect to the canonical inclusion $\partial\Delta^1 \hookrightarrow \Delta^1$.*

Proof. apparently this follows from ANOTHER theorem idk ■

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